The Spectral Shortcut Theorem: A Gradient-Geometric Explanation for Failure Modes of Joint Spectral-Spatial Deep Learning

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Abstract

[TODO: Write abstract last, after Phase 7.]

1 Introduction

The integration of spectroscopic imaging with deep learning has produced a class of compositional models that has become the de facto standard across infrared tissue classification, hyperspectral remote sensing, Raman imaging, and mass spectrometry imaging. A canonical pipeline processes a hyperspectral input $X \in \mathbb{R}^{H \times W \times S}$ in two stages: a per-pixel spectral reduction $f_{\boldsymbol{\theta}} : \mathbb{R}^S \to \mathbb{R}^K$ followed by a spatial model $g_{\boldsymbol{\phi}} : \mathbb{R}^{K \times H \times W} \to \mathbb{R}^{N_{\text{cls}} \times H \times W}$. End-to-end joint training of this pipeline is overwhelmingly the default — adopted by foundational work in FTIR histology Berisha et al. [2019], recent state-of-the-art spatial-spectral ViTs for prostate tissue classification O'Leary et al. [2026], and essentially all hyperspectral remote sensing systems.

An empirical paradox. Despite the prevalence of this paradigm, an unexpected and persistent pattern has emerged across multiple subfields. Frozen pretrained spectral encoders dramatically outperform end-to-end joint training on held-out test data. In our motivating tissue classification setting, a learned-from-scratch linear $f_{\boldsymbol{\theta}}$ trained end-to-end achieves 0.675 test F1; freezing a pretrained $f_{\boldsymbol{\theta}}$ yields 0.84–0.95 test F1. Most strikingly, replacing $f_{\boldsymbol{\theta}}$ with *frozen random weights* yields 0.70 test F1 — outperforming the joint trained model. Recent independent observations O'Leary et al. [2026], Berisha et al. [2019] have noted, in different terms, that aggressive spectral compression to 16 features has negligible impact on spatial-spectral neural network performance, and have concluded that "tissue classification only needs a small set of spectral features."

We argue the opposite. The reason end-to-end joint training fails is not that spectral information is unimportant. It is that *end-to-end optimization fails to extract that information*, even when the spectral signal is rich and discriminative. The problem is in the training procedure, not the data.

This paper. We provide a rigorous explanation for the failure of joint spectral-spatial training, grounded in the geometry of the loss landscape and the dynamics of stochastic gradient descent on compositional architectures with asymmetric capacity. Two main theorems form our contribution.

- **Theorem 14 (Capacity-Induced Module Curvature Disparity).** We prove that the ratio of the spatial to the spectral Gauss-Newton block's top eigenvalue grows linearly with the spatial module's width M (equivalently, as $\sqrt{C_g/C_f}$ for architecture families with $C_g = \Theta(M^2)$): as the spatial module widens, curvature concentrates in it while the spectral block's curvature stays capped by a width-independent constant. The proof is elementary: an explicit witness direction in the classifier-head weights forces the width-linear growth, and an operator-norm bound caps the spectral block; the rate matches the mean-field law of Karakida et al. Karakida et al. [2019] in the model class where the latter applies.

- **Theorem 27 (Finite-Time Spectral Starvation).** Under an explicit shortcut-fitting hypothesis compatible with cross-entropy dynamics, the spectral module's displacement from initialization by the time the training residual reaches level δ is at most $\mathcal{O}(\varepsilon \log(1/\delta))$, where ε is the ratio of the (capped) spectral gradient scale to the shortcut fitting rate. As spatial capacity grows, the shortcut fits faster while the spectral gradient stays capped, so the spectral module is pinned near initialization for the entire useful duration of training. The formulation is finite-time by design: for unregularized cross-entropy on separable data no finite stationary point exists Soudry et al. [2018], and the starvation phenomenon is a training-time effect, not an asymptotic one.

A real-time diagnostic. The pathology described by Theorems 14 and 27 is invisible to standard loss curves — training and validation loss both decay smoothly while the spectral module is silently starved. To make the pathology observable in practice we introduce the *Effective Gradient Ratio* (EGR), a scalar diagnostic computable inside any training loop:

$$\mathrm{EGR}(t) = \frac{\|\nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}_t, \boldsymbol{\phi}_t)\|}{\|\nabla_{\boldsymbol{\phi}} \mathcal{L}(\boldsymbol{\theta}_t, \boldsymbol{\phi}_t)\|}.$$

Proposition 33 establishes that the spectral gradient norm—the EGR's numerator—decays polynomially while the loss is being fitted. Empirically, the depth of the EGR collapse scales monotonically with capacity ratio, providing a quantitative, training-time observable for gradient starvation.

Prescription. The theory yields a direct prescription: freeze the spectral module, or inject a fixed residual baseline. Freezing eliminates the timescale gap by removing $\boldsymbol{\theta}$ from the optimization entirely. A residual baseline ($Z = f_{\boldsymbol{\theta}}(X) + h(X)$ for a fixed h, e.g. PCA) is a softer intervention that breaks the shortcut without precluding spectral fine-tuning. We discuss both in Section 9.

Universality. Although our motivating application is infrared tissue classification, the mechanism we identify applies to any compositional architecture with capacity asymmetry between sequential modules. We verify in Section 7 that the same pattern manifests across linear, CNN, and Vision Transformer spatial models on synthetic hyperspectral problems, indicating the phenomenon is architecture- agnostic. We expect the same mechanism to govern frozen-encoder practices in Raman spectroscopy, mass spectrometry imaging, NIR analysis, and beyond.

Relation to prior work. The closest prior work is in multimodal learning, where Wang et al. Wang et al. [2020] and Peng et al. Peng et al. [2022] identified "modality competition" and proposed gradient blending and modulation interventions. Our setting is unimodal (a single hyperspectral input) but the mechanism is structurally analogous: the spatial and spectral subspaces of the input act as competing "modalities," with the spatial module monopolizing the gradient budget. We discuss this connection further in Section 2.

Outline. Section 2 reviews related work. Section 3 establishes notation. Section 4 proves Theorem 14. Section 5 proves Theorem 27. Section 6 formalizes the EGR diagnostic. Section 7 validates both theorems and the diagnostic on synthetic spectral-spatial problems with linear, CNN, and ViT spatial models. Section 8 briefly connects the synthetic findings to real FTIR/QCL tissue classification data. Section 9 discusses prescriptions, limitations, and open problems.

2 Related Work

Our analysis sits at the intersection of four lines of research: the implicit bias and gradient dynamics of cross-entropy training; the geometry of deep-network loss surfaces; multimodal gradient competition; and spectral-spatial deep learning for biomedical and remote-sensing applications. We survey each in turn.

2.1 Gradient starvation and implicit bias

Cross-entropy minimization is known to induce structural biases in the learned representation. Soudry et al. Soudry et al. [2018] proved that on linearly separable data, gradient descent on logistic loss converges to the maximum-margin classifier. Gunasekar et al. Gunasekar et al. [2017] and Lyu and Li Lyu and Li [2020] extended this picture to deep homogeneous networks. These results establish that gradient-based training selects *which* solution among many that fit the training data.

Pezeshki et al. Pezeshki et al. [2021] identified a distinct phenomenon termed *gradient starvation*: in the Neural Tangent Kernel regime, dominant features in the input rapidly saturate the cross-entropy loss, after which weaker but task-relevant features receive negligible gradient signal. Our Theorem 27 adapts this framework to compositional architectures, where the "dominant features" are introduced not by the input distribution but by an architectural capacity asymmetry between modules. The result is a module-level starvation rather than a feature-level one.

2.2 Two-timescale stochastic approximation

The classical machinery for analyzing coupled gradient flows with disparate timescales is Borkar's stochastic approximation framework Borkar [2008]. Heusel et al. Heusel et al. [2017] adapted this to neural network training by introducing the Two Time-Scale Update Rule for GANs, proving convergence to a local Nash equilibrium under explicit step-size disparity. Our setting differs structurally: the timescale separation in our compositional model is not imposed by the optimizer (the same learning rate is applied to both modules) but emerges *implicitly* from the Hessian eigenvalue gap quantified by Theorem 14.

2.3 Hessian geometry of deep networks

The eigenspectrum of the neural-network Hessian has been characterized empirically and theoretically. Sagun et al. Sagun et al. [2017] identified a bulk-plus-outliers structure for overparametrized networks. Pennington and Bahri Pennington and Bahri [2017] developed a random matrix theory analysis that predicts spectral densities under Gaussian assumptions. Karakida et al. Karakida et al. [2019] derived closed-form mean-field expressions for the leading eigenvalue of the Fisher Information Matrix, showing it scales linearly with the layer width. Theorem 14 works blockwise on the joint Gauss–Newton matrix of a compositional model: a width-linear lower bound on the spatial block's top eigenvalue is combined with a width-independent operator-norm cap on the spectral block, yielding a width-scaling law for the module curvature disparity $\mathcal{D}_{\mathrm{curv}}$.

2.4 Multimodal gradient competition

The phenomenon most structurally analogous to ours is gradient competition in multimodal learning. Wang, Tran, and Feiszli Wang et al. [2020] showed empirically that joint training of multimodal classification networks underperforms unimodal training because different modalities overfit and generalize at different rates. They proposed Gradient Blending to dynamically reweight modality-specific losses. Peng et al. Peng et al. [2022] identified “modality laziness” — a tendency for one modality to dominate training, leaving the other under-optimized — and proposed On-the-fly Gradient Modulation. These works are empirical; our contribution adapts and extends their framework to the unimodal *compositional* setting and provides a rigorous geometric foundation. The recent biomedical application of gradient blending to multimodal sarcoma prognosis Bozzo et al. [2024] confirms the mechanism operates in medical settings, but is restricted to explicit dual-modality (MRI + clinical) architectures, not the single-modality compositional pipelines we consider.

2.5 Spectral-spatial deep learning

Spectral-spatial deep learning has been an active area for over a decade, primarily in two communities: biomedical infrared spectroscopy and hyperspectral remote sensing. Berisha et al. Berisha et al. [2019] established the benefit of combining spectral and spatial features for FTIR histology and observed that some tissue types could be classified from spatial features alone, with “minimal chemical information” required — a foreshadowing of the spatial shortcut phenomenon we diagnose. More recently, O’Leary et al. O’Leary et al. [2026] systematically compared spatial-spectral architectures for prostate cancer classification in infrared spectroscopy and reported the striking finding that compressing the spectral dimension to as few as 16 features had negligible effect on all models tested. The authors interpreted this as evidence that tissue classification requires only a limited spectral signature. Our Theorem 27 provides an alternative explanation: end-to-end training fails to extract the spectral signature even when it is present.

The conceptually nearest work in biomedical imaging is the recent characterization of shortcut learning in histopathology by Dawood and Minhas et al. Dawood et al. [2026], who showed that AI models predicting molecular biomarkers from H&E images exploit spurious correlations with clinicopathological features rather than learning genuine biological signal. Our paper extends the shortcut-learning framework into a new modality (spectroscopic imaging) with a different mechanism (gradient competition driven by architectural asymmetry rather than confounder leakage).

2.6 Frozen and random features

Frozen random features have a long history in machine learning. Saxe et al. Saxe et al. [2011] showed that convolutional networks with random (untrained) filters achieve competitive performance on standard benchmarks. Rahimi and Recht Rahimi and Recht [2007] established the theoretical foundations of random features for kernel methods. More recently, Coil and Cheney Coil and Cheney [2025] showed that freezing layers acts as implicit regularization in overparametrized networks. These works establish that frozen random features are not an oddity but a principled choice. Our contribution is to identify the specific mechanism — gradient starvation under capacity asymmetry — under which freezing is not merely competitive but *strictly preferable* to joint training.

3 Setup and Preliminaries

We analyze a generic two-module compositional model. While our motivating application is infrared spectroscopic imaging for tissue classification, the framework applies to any pipeline that first reduces a per-element spectral signal and then applies a high-capacity spatial model. We deliberately keep the notation domain-agnostic.

3.1 The compositional model

Let $X \in \mathbb{R}^{H\times W\times S}$ denote a hyperspectral input image, where H and W are spatial dimensions and S is the spectral dimension.[1] We write the per-pixel spectrum at location (h, w) as $x_{h,w} \in \mathbb{R}^S$.

The model consists of two modules applied sequentially:

1. A *spectral reduction* $f_{\boldsymbol{\theta}} : \mathbb{R}^S \to \mathbb{R}^K$ applied independently to every pixel, producing a feature map $Z \in \mathbb{R}^{K\times H\times W}$ with $Z_{:,h,w} = f_{\boldsymbol{\theta}}(x_{h,w})$.
2. A *spatial model* $g_{\boldsymbol{\phi}} : \mathbb{R}^{K\times H\times W} \to \mathbb{R}^{N_{\text{cls}}\times H\times W}$ producing per-pixel class logits $\hat{y} = g_{\boldsymbol{\phi}}(Z)$.

We denote the composed model $F_{\boldsymbol{\theta},\boldsymbol{\phi}}(X) = g_{\boldsymbol{\phi}}(f_{\boldsymbol{\theta}}(X))$ and the parameter counts $C_f := \dim(\boldsymbol{\theta})$ and $C_g := \dim(\boldsymbol{\phi})$.

Assumption 1 (Linear spectral reduction)**.** $f_{\boldsymbol{\theta}}$ is linear in $\boldsymbol{\theta}$: there exists $W \in \mathbb{R}^{S\times K}$ with $\boldsymbol{\theta} = \mathrm{vec}(W)$ and $f_{\boldsymbol{\theta}}(x_{h,w}) = W^\top x_{h,w}$.

Remark 2. Assumption 1 captures the canonical case used widely in spectroscopy pipelines, including the FTIR and QCL tissue classification work we draw motivation from. It is also the cleanest case in which the Hessian of $\mathcal{L}$ admits closed-form block analysis. We discuss extensions to shallow nonlinear $f_{\boldsymbol{\theta}}$ in Section 9 and verify empirically in Section 7 that the theorem's scaling persists when $f_{\boldsymbol{\theta}}$ is replaced by a small MLP.

Assumption 3 (Spatial model regularity)**.** $g_{\boldsymbol{\phi}}$ is twice continuously differentiable jointly in $(\boldsymbol{\phi}, Z)$ with L-Lipschitz gradients. For piecewise-smooth models (e.g. ReLU-type activations) this is relaxed: the gradient flow is interpreted in the almost-everywhere sense of Section 5, and no quantitative bound in this paper consumes second derivatives.

Assumption 4 (Bounded input second moment)**.** Define the *mean empirical input Gram*

$$\Sigma_X \;:=\; \frac{1}{N}\sum_{n=1}^{N} x_n x_n^\top, \qquad N \;:=\; N_{\text{data}}HW, \tag{1}$$

with x_n enumerating the per-pixel spectra of all images. We assume $\lambda_{\max}(\Sigma_X) \le \Lambda_X < \infty$, with Λ_X uniform over the datasets under consideration. Every constant below is stated in terms of $\lambda_{\max}(\Sigma_X)$ (or, where noted, $\mathrm{tr}(\Sigma_X)$) and is therefore independent of the dataset size.

Remark 5 (Population motivation—not used in proofs). For a fixed finite dataset, $\lambda_{\max}(\Sigma_X)$ is finite automatically, and this empirical quantity is all the proofs consume; the assumption has content only along sequences of growing datasets. If the pixel spectra are modelled as sub-Gaussian draws from a population with second moment $\Sigma = \mathbb{E}[xx^\top]$ that are independent (or sufficiently weakly dependent) *across pixels*, then $\lambda_{\max}(\Sigma_X) \to \lambda_{\max}(\Sigma)$ and the uniform bound holds with Λ_X near $\lambda_{\max}(\Sigma)$. Marginal sub-Gaussianity alone does *not* suffice: neighbouring pixels of a tissue image are strongly correlated, and in the extreme case of perfectly duplicated pixels Σ_X stays bounded but never concentrates around Σ. This is why we assume the empirical bound directly rather than derive it from population conditions. Well-conditionedness of Σ (normalized inputs) is likewise invoked only in the discussion of classical remedies (Section 4.5), never in a proof.

[1] Multi-channel inputs (e.g. raw absorbance together with first and second derivatives) are handled by concatenating channels along the spectral dimension; this changes only the input size and leaves the analysis unaffected.

Assumption 6 (Capacity gap)**.** $C_g/C_f \to \infty$.

Assumption 6 is not used in any proof: it is the structural regime in which our theorems become *informative* rather than merely true, and we state it to delimit scope. Empirically it is satisfied with room to spare in modern spectral-spatial architectures: in our motivating example (Section 8, Table 2) the ratio is $C_g/C_f \approx 300$, roughly two orders of magnitude.

Assumption 7 (Bounded input Jacobian of the classifier)**.** There exists a constant $\widetilde{L} > 0$ depending on the depth of $g_{\boldsymbol{\phi}}$, the bottleneck dimension K, the number of classes N_{cls}, and the data distribution — but *not* on the characteristic width M or the total spatial parameter count C_g — such that

$$\sup_{\boldsymbol{\phi}\in\Phi_{\mathrm{reg}},\ Z\in\mathcal{Z}_{\mathrm{reg}}} \left\| \frac{\partial g_{\boldsymbol{\phi}}(Z)}{\partial Z} \right\|_{\mathrm{op}} \leq \widetilde{L},$$

uniformly over the training-time parameter regime Φ_{reg} and the induced feature domain $\mathcal{Z}_{\mathrm{reg}}$.

Remark 8 (Why Assumption 7 is separate from Assumption 3). Assumption 3 is a *regularity* statement: smoothness of $g_{\boldsymbol{\phi}}$ jointly in parameters and features, which is what well-posedness of the joint flow—and the chain rule along it—require ($\boldsymbol{\theta}$ enters the loss only through Z, so differentiability of $t \mapsto \mathcal{L}(t)$ needs the Z-derivative too). Assumption 7 is a *quantitative, width-uniform bound* on the first Z-derivative—a different kind of statement with a different constant: smoothness alone implies no uniform bound. Assumption 3's role is to keep the gradient flow (7) well posed; the quantitative bounds of Theorems 14 and 27 use only Assumption 7.

What random-matrix theory does and does not supply here deserves care. For a random weight matrix whose *two dimensions grow proportionally*, variance-scaled initialization gives an operator norm that is $\Theta(1)$ in width: this is the classical largest-singular-value edge (Yin, Bai, and Krishnaiah Yin et al. [1988]; Bai and Yin Bai and Yin [1993]; non-asymptotically, Vershynin Vershynin [2018], Thm. 4.4.5; Marchenko–Pastur Marčenko and Pastur [1967] describes the corresponding bulk spectrum). It does *not* apply to the layer reading the bottleneck: a Kaiming-initialized expansion matrix $W_1 \in \mathbb{R}^{M\times K}$ with fixed K and entry variance σ_w^2/K has $\|W_1\|_{\mathrm{op}} = \Theta(\sqrt{M/K})$, which *diverges* with width, so a per-layer product-of-operator-norms argument inherits this factor. The boundedness of the composed Jacobian $\partial g_{\boldsymbol{\phi}}/\partial Z$ is instead a cancellation effect across consecutive variance-scaled layers: the $1/\sqrt{M}$ entry scaling of the *next* layer offsets the $\sqrt{M}$ growth, because the top directions of independent random factors are generically misaligned. We do not prove this in general—uniform boundedness is exactly what Assumption 7 postulates—but we prove it for a concrete two-layer ReLU head in Proposition 38 (supplement), where $\mathbb{E}\,\|\partial\hat{y}/\partial z\|_F^2 = \mathcal{O}(1)$ uniformly in width while the *squared* naive product bound $\|W_2\|_{\mathrm{op}}^2\|W_1\|_{\mathrm{op}}^2$ would give $\Theta(M/K)$. Under the maximal-update parameterization (μP), Yang and Hu Yang and Hu [2021] establish coordinate-wise $\Theta(1)$ stability of activations throughout training in the infinite-width limit; we take the corresponding operator-norm stability as a modeling assumption motivated by that result, not as a consequence of it. Under standard-parameterization training with weight decay $\lambda > 0$ and bounded-norm iterates, the per-layer product-of-operator-norms bound gives a *finite* $\widetilde{L}$ over any training horizon—but not a width-uniform one, since per-layer norms at Kaiming scale grow like $\sqrt{M}$ and weight decay holds them near that scale rather than below it. Width-uniformity is therefore the modeling content of Assumption 7, supported at initialization by the composition-cancellation effect above (proved in expectation for the two-layer head) rather than derived from norm control. Softmax self-attention is Lipschitz only on bounded inputs Kim et al. [2021], so $\mathcal{Z}_{\mathrm{reg}}$ should be taken compact whenever $g_{\boldsymbol{\phi}}$ uses attention. Pre-attention LayerNorm (standard practice) supports this: it places each token on a sphere whose radius is set by the learned affine gains, so attention inputs stay bounded as long as those gains do, i.e. within the bounded parameter regime Φ_{reg}. We impose the compactness as part of the assumption rather than derive it.

3.2 Loss and gradients

We train with per-pixel cross-entropy, averaged over all N pixel positions of the dataset:

$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}) = \frac{1}{N} \sum_{n=1}^{N} \mathrm{CE}(\hat{y}_n,\ y_n)\,, \qquad \hat{y}_n = F_{\boldsymbol{\theta},\boldsymbol{\phi}}(X)_n, \tag{2}$$

where $y_n \in \{1, \ldots, N_{\mathrm{cls}}\}$ is the ground-truth class. Cross-entropy plays an essential role in Section 5: its gradient with respect to the logits $\hat{y}$ is the prediction residual $r_n = \mathrm{softmax}(\hat{y}_n) - y_{n,\mathrm{onehot}}$, which vanishes exponentially in the logit margin as predictions saturate—and hence, on separable data, polynomially in training time (Section 5). We discuss other loss functions sharing this vanishing-residual property in Section 9.

Normalization convention. Consistently with (2), we collect the per-pixel quantities into N-normalized stacked objects: the residual vector $r := N^{-1/2}(r_1, \ldots, r_N)$, so that $\|r\|$ is the root-mean-square per-pixel residual, and the *logit Jacobian*

$$J_\bullet \ := \ N^{-1/2}\, \frac{\partial(\hat{y}_1, \ldots, \hat{y}_N)}{\partial \bullet}, \qquad \bullet \in \{\boldsymbol{\theta}, \boldsymbol{\phi}\}, \tag{3}$$

so that $\nabla_\bullet \mathcal{L} = J_\bullet^\top r$ exactly. Two Jacobian-like objects appear in the literature and we fix both here: $J_\bullet$ above (logit Jacobian), and the *residual Jacobian* $J_\bullet^{\mathrm{res}} := \partial r/\partial \bullet = H_\tau J_\bullet$, where $H_\tau := \mathrm{diag}(p) - pp^\top$ is the softmax Jacobian (blockwise over pixels). Since $0 \preceq H_\tau \preceq I$, any operator-norm bound on $J_\bullet$ applies a fortiori to $J_\bullet^{\mathrm{res}}$. The intermediate factor $\partial Z/\partial \boldsymbol{\theta}$ carries the same $N^{-1/2}$ stacking, so that the chain rule composes as $J_{\boldsymbol{\theta}} = (\partial \hat{y}/\partial Z) \cdot (\partial Z/\partial \boldsymbol{\theta})$ with the unnormalized middle factor — which is the object Assumption 7 bounds. All Gram matrices, and Σ_X in (1), use this same $1/N$ normalization, so no constant in this paper depends on the dataset size.

The gradient of $\mathcal{L}$ with respect to the spatial parameters is computed by standard back-propagation. The gradient with respect to the spectral parameters $\boldsymbol{\theta}$ admits the chain-rule decomposition

$$\nabla_{\boldsymbol{\theta}} \mathcal{L} = \underbrace{r}_{\text{residual}} \cdot \underbrace{\frac{\partial \hat{y}}{\partial Z}}_{\text{spatial Jacobian}} \cdot \underbrace{\frac{\partial Z}{\partial \boldsymbol{\theta}}}_{\text{input}} = J_{\boldsymbol{\theta}}^\top r, \tag{4}$$

with all three factors in the normalization fixed above (the per-pixel loss gradients $\partial \mathcal{L}/\partial \hat{y}_n = r_n/N$ are absorbed into the $N^{-1/2}$-scaled r and $\partial Z/\partial \boldsymbol{\theta}$). The softmax enters through the residual itself; the matrix H_τ appears explicitly only when differentiating r, as in J^{res} above and in the Gauss-Newton blocks below. The three factors play distinct roles in our analysis. The residual is small when predictions are confident. The spatial Jacobian depends on the current state of $\boldsymbol{\phi}$ and thus changes throughout training. The final factor is the input itself and is stationary. The product structure implies that if the residual decays to zero, the gradient to $\boldsymbol{\theta}$ vanishes regardless of the other two factors. This observation underlies Theorem 27.

Remark 9 (How much of this depends on cross-entropy?). The two theorems depend on the loss to different degrees. Theorem 14 is essentially loss-agnostic: the softmax Jacobian H_τ enters its proof only through the upper bound $H_\tau \preceq I$ and through the per-pixel weights $p_{\min}^{(n)}$ that the head hypothesis carries—both contribute constants and no M-dependence. Under squared error the inner matrix is the identity and the argument simplifies, giving the same width-linear disparity. Theorem 27 is more loss-specific: the polynomial residual envelope of Assumption 24 is the characteristic behavior of logistic-type losses on separable data (Soudry et al. Soudry et al. [2018]), whereas squared error on a well-conditioned target admits a faster (exponential) envelope. As noted after Assumption 24, a faster envelope only tightens the displacement bound; the mechanism—a fast module driving the residual down and starving a slow one through the shared factor—is not specific to cross-entropy, though the $\log(1/\delta)$ rate is.

3.3 The block Hessian

The joint Hessian of $\mathcal{L}$ decomposes into blocks aligned with the two parameter groups:

$$H = \nabla^2 \mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}) = \begin{pmatrix} H_{\boldsymbol{\theta\theta}} & H_{\boldsymbol{\theta\phi}} \\ H_{\boldsymbol{\phi\theta}} & H_{\boldsymbol{\phi\phi}} \end{pmatrix}, \tag{5}$$

with $H_{\boldsymbol{\theta\theta}} \in \mathbb{R}^{C_f \times C_f}$, $H_{\boldsymbol{\phi\phi}} \in \mathbb{R}^{C_g \times C_g}$, and cross blocks of appropriate sizes. Theorem 14 bounds the *ratio* of the two diagonal blocks' top eigenvalues (Definition 12) in terms of the spatial width, by bounding each block separately—not the condition number of H, which is a different and, here, degenerate quantity (Remark 13).

For curvature analysis we work with the generalized Gauss-Newton matrix, which for softmax cross-entropy is

$$G \;=\; J^\top H_\tau J, \qquad J = [\, J_{\boldsymbol{\theta}},\; J_{\boldsymbol{\phi}} \,], \tag{6}$$

with J the (N-normalized) logit Jacobian of (3) and H_τ the softmax Jacobian (Botev et al. Botev et al. [2017]). G is the positive semidefinite part of H that survives at interpolation, and $H = G$ exactly when the residual vanishes. Theorem 14 is stated for the diagonal blocks $G_{\boldsymbol{\theta\theta}} = J_{\boldsymbol{\theta}}^\top H_\tau J_{\boldsymbol{\theta}}$ and $G_{\boldsymbol{\phi\phi}} = J_{\boldsymbol{\phi}}^\top H_\tau J_{\boldsymbol{\phi}}$. The residual-Gram variant $J^{\mathrm{res}\top} J^{\mathrm{res}} = J^\top H_\tau^2 J$ obeys the same upper bounds unconditionally ($H_\tau^2 \preceq I$) and the same lower bound under a correspondingly weighted form of the head hypothesis (supplement Section A.1), so the results transfer across conventions with only the softmax weighting of the constants changing.

3.4 The Effective Gradient Ratio

We introduce a real-time diagnostic that tracks the asymmetry in gradient flow between the two modules.

Definition 10 (Effective Gradient Ratio)**.** At training step t with parameters $(\boldsymbol{\theta}_t, \boldsymbol{\phi}_t)$, the Effective Gradient Ratio is

$$\mathrm{EGR}(t) \;:=\; \frac{\|\nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}_t, \boldsymbol{\phi}_t)\|}{\|\nabla_{\boldsymbol{\phi}} \mathcal{L}(\boldsymbol{\theta}_t, \boldsymbol{\phi}_t)\|}.$$

Remark 11. EGR is a unitless quantity directly computable inside any training loop with negligible overhead. Section 6 bounds the decay of its numerator—the spectral gradient norm—by $B_T C/(1+\mu t)$ under the hypotheses of Theorem 27, characterizes the ratio's trajectory empirically, and proposes a threshold criterion for detecting gradient starvation onset.

3.5 Notational conventions

Throughout, $\|\cdot\|$ denotes the Euclidean (ℓ_2) norm for vectors and the spectral norm for matrices; $\lambda_{\max}(\cdot)$ and $\lambda_{\min}(\cdot)$ denote the largest and smallest eigenvalues of a symmetric matrix; $\kappa(A) = \lambda_{\max}(A)/\lambda_{\min}(A)$ its condition number. We write $A \succeq B$ to mean $A - B$ is positive semidefinite. The symbol $\Omega(\cdot)$ is used in the standard asymptotic sense.

Definition 12 (Module curvature disparity)**.** For the Gauss-Newton matrix G of (6) with diagonal blocks $G_{\boldsymbol{\phi\phi}}, G_{\boldsymbol{\theta\theta}}$ indexed as in (5), the *module curvature disparity* is

$$\mathcal{D}_{\mathrm{curv}}(G) \;:=\; \frac{\lambda_{\max}(G_{\boldsymbol{\phi\phi}})}{\lambda_{\max}(G_{\boldsymbol{\theta\theta}})},$$

the ratio of the spatial block's top eigenvalue to the spectral block's, with the convention $\mathcal{D}_{\mathrm{curv}} := +\infty$ when the spectral block vanishes ($\lambda_{\max}(G_{\boldsymbol{\theta\theta}}) = 0$, a maximal disparity; Assumption 7 caps the spectral block from above only, so the degenerate case is not excluded, and lower bounds on $\mathcal{D}_{\mathrm{curv}}$ are unaffected by it). Unless stated otherwise, $\mathcal{D}_{\mathrm{curv}}$ is evaluated at random initialization $(\boldsymbol{\theta}_0, \boldsymbol{\phi}_0)$, which is where Theorem 14 bounds it and where Experiment 1.1 measures it.

Remark 13 (Why disparity, not condition number). Theorem 14 is a purely geometric statement about how curvature concentrates between the two modules as the spatial network widens. It is not a joint-Hessian condition-number bound: the joint Gauss-Newton G is rank-deficient (parameter directions orthogonal to the classifier's range have exactly zero curvature), so $\kappa(G) = \lambda_{\max}(G)/\lambda_{\min}(G)$ is trivially infinite. The disparity $\mathcal{D}_{\text{curv}}$ is well-defined regardless of rank structure, uses only well-behaved operator-norm quantities, and is exactly the ratio measured in Experiment 1.1. Its growth with spatial capacity is the geometric mechanism that motivates the two-timescale dynamical analysis of Theorem 27, but the dynamical claim requires its own hypotheses and does not follow from $\mathcal{D}_{\text{curv}}$ alone.

4 Theorem 1: Capacity-Induced Module Curvature Disparity

This section establishes our first main result: the module curvature disparity $\mathcal{D}_{\text{curv}}$ (Definition 12) grows with the spatial module's width. The result is the geometric foundation for the dynamical analysis of Section 5: it identifies a capacity-dependent mechanism capable of generating disparate module learning rates. The dynamical starvation claim itself is stated separately in Theorem 27, under its own explicit hypotheses.

4.1 Main result

Theorem 14 (Capacity-Induced Module Curvature Disparity). *Consider the compositional model $F_{\boldsymbol{\theta},\boldsymbol{\phi}} = g_{\boldsymbol{\phi}} \circ f_{\boldsymbol{\theta}}$ with fixed-dimensional linear spectral module $f_{\boldsymbol{\theta}}$ (Assumption 1) and a family of spatial networks $g_{\boldsymbol{\phi}}^{(M)}$ indexed by characteristic width M (channels for a CNN, embedding dimension for a ViT, hidden width for an MLP). Under Assumptions 1, 4, and 7, together with the head hypothesis of Lemma 16 below (a dense classifier head over penultimate features of dimension $\Theta(M)$ that are correlated across inputs at initialization), there exist constants $c_1, c_2 > 0$ depending on the data distribution, the initialization scheme, the bottleneck dimension K, the input spectral dimension S, and the architecture class of $g_{\boldsymbol{\phi}}$ (depth, activation, head dimensions)—but not on M or on the spatial parameter count C_g—such that*

$$\mathcal{D}_{\text{curv}}^{(M)} \;=\; \frac{\lambda_{\max}(G_{\boldsymbol{\phi}\boldsymbol{\phi}}^{(M)})}{\lambda_{\max}(G_{\boldsymbol{\theta}\boldsymbol{\theta}})} \;\geq\; c_1 \cdot M - c_2$$

in expectation over random Kaiming (variance-scaled Gaussian) initialization. Equivalently, for architecture families with $C_g = \Theta(M^2)$,

$$\mathcal{D}_{\text{curv}}^{(M)} \;\geq\; c_1' \cdot \sqrt{C_g/C_f} - c_2.$$

Remark 15 (Interpretation). Theorem 14 states that as the spatial network widens, its top Gauss-Newton eigenvalue grows without bound while the spectral module's top Gauss-Newton eigenvalue remains capped by a C_g-independent constant. Curvature increasingly concentrates in the downstream module. This is a purely geometric fact about the Hessian; the dynamical consequence—that joint training pins $\boldsymbol{\theta}$ near its initialization—is stated separately in Theorem 27, under its own hypotheses. The disparity measures the downstream module's *readiness to move*—the loss curvature its parameters command at initialization—not the usefulness of its features: it is present before any feature has been learned, and the width-linear spike can be exhibited even for inputs carrying no information (Remark 40, supplement). That the geometry favours the spatial module *independently* of feature quality is precisely what makes it a shortcut hazard.

The proof decomposes the disparity bound into separate bounds on the top eigenvalues of the two diagonal blocks. We state the supporting lemmas before presenting the proof.

4.2 Supporting lemmas

Lemma 16 (Spatial block scaling)**.** *Let M denote the characteristic width of $g_{\boldsymbol{\phi}}$ (channel count for a CNN, embedding dimension for a ViT, hidden layer width for an MLP), let $N_{\mathrm{cls}} \geq 2$, and suppose $g_{\boldsymbol{\phi}}$ ends in a dense classifier head over penultimate features $h_n \in \mathbb{R}^{M_h}$ with $M_h = \Theta(M)$ whose softmax-weighted mean Gram $S_h^p := N^{-1} \sum_n p_{\min}^{(n)} h_n h_n^\top$ (Equation (11)) satisfies*

$$\lambda_{\max}(S_h^p) \;\geq\; q_h M_h \quad \textit{with probability } \geq 1 - \delta_h$$

over the random initialization—where $p_{\min}^{(n)}$ denotes the smallest softmax probability at pixel n—for constants $q_h > 0$ and $\delta_h < 1$ independent of M (the head hypothesis*: the features carry a common direction of $\Theta(M)$ energy across inputs, on pixels whose initial predictions are non-degenerate. It is a cross-input* correlation *condition, not a per-vector norm condition, and is closely related to the quantity κ_2 carrying the width-linear term in Karakida et al.'s law—see the supplement, where a fully proved instance is also given). Then the spatial Gauss-Newton block of (6) satisfies*

$$\lambda_{\max}(G_{\boldsymbol{\phi\phi}}) \;=\; \Omega(M) \qquad \begin{array}{c}\textit{in expectation over random}\\ \textit{Kaiming (variance-scaled}\\ \textit{Gaussian) initialization.}\end{array}$$

The logit Gram $J_{\boldsymbol{\phi}}^\top J_{\boldsymbol{\phi}} \succeq G_{\boldsymbol{\phi\phi}}$ inherits the bound unconditionally; the residual Gram $J_{\boldsymbol{\phi}}^{\mathrm{res}\top} J_{\boldsymbol{\phi}}^{\mathrm{res}}$ satisfies it under the analogous hypothesis on the p^2-weighted Gram $S_h^{p^2}$ (supplement). For architecture families with $C_g = \Theta(M^2)$ (fixed depth with at least two consecutive width-proportional layers), equivalently $\lambda_{\max}(G_{\boldsymbol{\phi\phi}}) = \Omega(\sqrt{C_g})$.

Remark 17. Lemma 16 is proved in the supplement (Section A.1) by an explicit witness: the rank-one head-weight perturbation $V = c\,u^\top$, where c is a unit class contrast and u the dominant direction of the feature ensemble. Evaluating the curvature along V gives the deterministic inequality $\lambda_{\max}(G_{\boldsymbol{\phi\phi}}) \geq \lambda_{\max}(S_h^p)$, and the head hypothesis supplies $\lambda_{\max}(S_h^p) = \Omega(M)$. The width-linear law agrees with the mean-field analysis of Karakida et al. Karakida et al. [2019]: their Theorem 4 gives $\lambda_{\max} = \Theta(M)$ with sharp constant for proportional-width fully connected networks under squared loss, and their Theorem 1 shows the mean eigenvalue simultaneously shrinks as $\mathcal{O}(1/M)$—curvature concentrates in a vanishing fraction of directions as networks widen. The scaling variable is width, not parameter count: for architecture families with $C_g = \Theta(M^2)$, the equivalent parameter-count statement is $\lambda_{\max} = \Omega(\sqrt{C_g})$. The head hypothesis covers per-pixel dense heads on CNN and ViT modules as well; the sharpness of the width-linear rate is tested empirically in Section 7.

Lemma 18 (Spectral block operator-norm cap)**.** *Under Assumptions 1 (linear per-pixel $f_{\boldsymbol{\theta}}$), 4 (bounded input second moment), and 7 (bounded input Jacobian $\|\partial g_{\boldsymbol{\phi}}/\partial Z\|_{\mathrm{op}} \leq \widetilde{L}$ uniformly in C_g), the spectral logit Jacobian and Gauss-Newton block satisfy*

$$\lambda_{\max}(G_{\boldsymbol{\theta\theta}}) \;\leq\; \|J_{\boldsymbol{\theta}}\|_{\mathrm{op}}^2 \;\leq\; \widetilde{L}^2 \cdot \lambda_{\max}(\Sigma_X) \;=:\; C_{\boldsymbol{\theta}},$$

where $C_{\boldsymbol{\theta}}$ depends only on the classifier's input Lipschitz constant $\widetilde{L}$ and the top eigenvalue of the mean per-pixel input Gram Σ_X of (1)—not on the spatial width M, the spatial parameter count C_g, or the dataset size N. The same bound holds for the residual-Gram convention, since $H_\tau^2 \preceq H_\tau \preceq I$.

Remark 19. Lemma 18 follows from the chain-rule identity $J_{\boldsymbol{\theta}} = J_Z \cdot (\partial Z/\partial \boldsymbol{\theta})$ for the logit Jacobian, combined with $\|J_Z\|_{\mathrm{op}} \leq \widetilde{L}$ (Assumption 7) and the exact form of $\partial Z/\partial \boldsymbol{\theta}$ under linearity of $f_{\boldsymbol{\theta}}$ (Assumption 1, giving the per-pixel factor $I_K \otimes x_n^\top$, whose N-normalized stacking has $\|\partial Z/\partial \boldsymbol{\theta}\|_{\mathrm{op}}^2 = \lambda_{\max}(\Sigma_X)$ exactly, by $I_K \otimes \Sigma_X$). Passing from the logit Jacobian to the

Gauss-Newton block or the residual Jacobian only inserts factors of $H_\tau \preceq I$ and can only decrease the bound. The bound is on the operator norm (equivalently, the top eigenvalue), which is well-behaved under rank-deficiency of the block—avoiding the smallest-eigenvalue and Kronecker-factorisation machinery discussed in Section A.1.

4.3 Proof of Theorem 14

Proof of Theorem 14. By Lemma 16, in expectation over random Kaiming (variance-scaled Gaussian) initialization,

$$\lambda_{\max}(G^{(M)}_{\boldsymbol{\phi}\boldsymbol{\phi}}) \;\geq\; c_\phi \cdot M$$

for a constant $c_\phi > 0$ independent of M (depending on the initialization scheme, the activation, the head dimensions, and the data distribution; supplement Section A.1). By Lemma 18,

$$\lambda_{\max}(G_{\boldsymbol{\theta}\boldsymbol{\theta}}) \;\leq\; \|J_{\boldsymbol{\theta}}\|^2_{\mathrm{op}} \;\leq\; C_{\boldsymbol{\theta}}$$

for a constant $C_{\boldsymbol{\theta}} > 0$ independent of M. The cap is deterministic—it holds for every realization of the initialization—so it may be divided through inside the expectation:

$$\mathbb{E}\Big[\mathcal{D}^{(M)}_{\mathrm{curv}}\Big] \;=\; \mathbb{E}\left[\frac{\lambda_{\max}(G^{(M)}_{\boldsymbol{\phi}\boldsymbol{\phi}})}{\lambda_{\max}(G_{\boldsymbol{\theta}\boldsymbol{\theta}})}\right] \;\geq\; \frac{\mathbb{E}\Big[\lambda_{\max}(G^{(M)}_{\boldsymbol{\phi}\boldsymbol{\phi}})\Big]}{C_{\boldsymbol{\theta}}} \;\geq\; \frac{c_\phi \cdot M}{C_{\boldsymbol{\theta}}} \;=\; c_1 \cdot M$$

with $c_1 := c_\phi / C_{\boldsymbol{\theta}}$. The witness route delivers the bound with $c_2 = 0$; the subtracted constant is retained in the statement so that sharper asymptotic constants, valid only for large M, may be substituted. Under $C_g = \Theta(M^2)$ (families with at least two consecutive width-proportional layers), $M = \Theta(\sqrt{C_g})$ and the ratio form $c_1' \cdot \sqrt{C_g/C_f}$ follows on multiplying and dividing by $\sqrt{C_f}$. This completes the proof. □

4.4 Implications

Theorem 14 establishes that curvature concentrates in the downstream spatial module as its width grows. We highlight two implications, and separate them explicitly from the dynamical consequences of Section 5.

Geometric mechanism, not a dynamical claim. $\mathcal{D}_{\mathrm{curv}} \gg 1$ identifies a *potential* for disparate module learning rates: the spatial Gauss-Newton block admits directions of arbitrarily large curvature while the spectral block does not. Whether joint gradient descent actually produces disparate effective learning rates depends on an additional dynamical hypothesis (that the shortcut pathway fits the residual at rate μ; Assumption 24), which Section 5 introduces and Experiment 1.2 tests. Theorem 14 does not, by itself, prove that $\boldsymbol{\theta}$ is pinned to initialization. In particular, $\mathcal{D}_{\mathrm{curv}}$ concerns the *top* of the spatial block's spectrum, while the fitting rate μ of Section 5 is governed by how fast the residual can be driven down—a property of other parts of the spatial spectrum. The two are linked only under spectral-shape assumptions we do not make; the theorems are deliberately independent, and Experiments 1.1–1.2 test them separately.

The disparity is invariant to shared step-size tuning. The disparity bound depends only on architectural parameters, not on the choice of optimizer or the shared step-size schedule—so no choice of shared learning rate changes the *geometry*; whether any shared-step-size method escapes the *dynamical* consequence is a separate question we treat as a conjecture below. Adaptive methods (Adam, RMSProp) rescale per-coordinate gradient magnitudes but do not alter the underlying eigenvalue disparity between the two blocks. Two-timescale optimizers with explicitly different learning rates per module (Heusel et al. Heusel et al. [2017]; Borkar 2008 Borkar [2008] Ch. 6) are the natural remedy at the optimizer level. We discuss this in Section 9.

4.5 Relation to classical ill-conditioning

Ill-conditioned optimization has classical causes with classical remedies, and it is natural to ask whether they apply here. They do not, because the disparity of Theorem 14 is driven by the *architecture*, not by the data or the parametrization of a single block.

Feature normalization. Scale asymmetry or collinearity among input features degrades the conditioning of Σ_X and is classically cured by standardizing (e.g. mean-centering) the inputs. Here the cure has a sharper, quantitative role: Lemma 18 places $\lambda_{\max}(\Sigma_X)$ in the spectral cap, so input transformations that deflate the top of the input spectrum *raise* the realized disparity at every width—normalization moves the constants. What no input transformation can touch is the width-scaling of the spatial block (Lemma 16): capacity moves the rate. On raw nonnegative spectra, whose second moment is dominated by the shared mean shape, the cap is large and the realized disparity correspondingly small. Whether an input transformation moves the *realized* disparity, however, depends on the architecture between $f_{\boldsymbol{\theta}}$ and the loss: internal normalization layers can absorb input shifts entirely (their Jacobians annihilate constant directions), in which case the effective regressor is already centered and the lever acts through the normalization choice, not the preprocessing. We measure exactly this on the production architecture in Section 8.

Weight decay. Adding $\alpha \|w\|^2$ to the loss shifts the Hessian by $2\alpha I$, lifting every eigenvalue by 2α and famously repairing classical condition numbers. But the shift acts on *both* blocks equally: the disparity becomes $(\lambda_{\max}(G_{\boldsymbol{\phi}\boldsymbol{\phi}}) + 2\alpha)/(\lambda_{\max}(G_{\boldsymbol{\theta}\boldsymbol{\theta}}) + 2\alpha)$, which remains $\Theta(M)$ for any fixed α as M grows. Repairing the disparity would require $2\alpha \sim \lambda_{\max}(G_{\boldsymbol{\phi}\boldsymbol{\phi}})$, a regularization strength that would dominate the data term and prevent fitting. Consistent with this, Experiment 1.6 finds that Pezeshki's Spectral Decoupling—an L_2 penalty on logits designed to combat saturation—does not reduce the joint-training collapse, while freezing does.

Adaptive optimizers. Adam and RMSProp precondition per-coordinate gradient magnitudes with a diagonal rescaling. The raw block spectra are optimizer-invariant, but the *dynamics* under a diagonal preconditioner P are governed by the preconditioned spectra of $P^{-1/2}GP^{-1/2}$ and need not respect the disparity: under Adam, per-coordinate steps are near-constant in gradient magnitude, so small spectral gradients do not imply small spectral movement. The gradient-flow statements of Section 5 are accordingly scoped to (stochastic) gradient descent; joint-vs-frozen gaps observed under Adam are evidence about the phenomenon, not applications of the theorem. Empirically, the canonical dynamics experiment (Experiment 1.2) is run under Adam and the joint-vs-frozen gap persists at every width; we treat the stronger claim—that no shared-learning-rate adaptive method can remove the pathology—as a conjecture rather than a theorem. Optimizers with explicitly different learning rates per module (two-timescale update rules; Heusel et al. Heusel et al. [2017]) are the natural remedy at the optimizer level and correspond, in the limit, to our freezing prescription.

Remark 20 (Parameterization scope). $\mathcal{D}_{\text{curv}}$ is not invariant to per-module reparameterization, and Theorem 14 is a statement about the standard variance-scaled (Kaiming) parameterization—the one used in the pipelines we analyze and in all our experiments. Under alternative parameterizations that rescale layers with width, such as μP Yang and Hu [2021], both the initialization magnitudes and the effective per-layer learning rates change, and the width scaling of the blocks would need to be re-derived; we do not claim the disparity law transfers. Our two uses of μP elsewhere are limited to a different role: as one route to keeping the input-Jacobian bound (Assumption 7) uniform in width during training. Whether principled reparameterization can neutralize the disparity—rather than merely re-expressing it—is an open question adjacent to the per-module learning-rate remedies discussed in Section 9.

Remark 21 (Small eigenvalues: irrelevance vs. starvation). A parameter direction can carry vanishing curvature for two very different reasons: (a) the direction genuinely does not affect predictions on the task at hand ("irrelevance"), in which case zero gradient is the correct outcome and the parameter should not be learned; or (b) the direction *would* affect predictions if the optimizer could reach it, but the compositional geometry buries the gradient signal behind the bottleneck ("starvation"). The Gauss-Newton bounds of Theorem 14 alone cannot distinguish between these interpretations: both produce C_g-independent small curvature in the spectral block.

The synthetic experiments in Section 7 are designed to distinguish these hypotheses by construction. In Experiment 1.3 we generate data where each of the two modalities alone carries information sufficient to solve the task (single-modality baselines reach ≈ 0.53 on a binary classification, above the 0.50 chance floor). Joint training on the same data does not merely under-use the spectral channel: it undershoots the weaker of the two single-modality baselines. The gap is modest in absolute terms on this synthetic problem—an important honest caveat—but its direction is inconsistent with interpretation (a), under which joint training would at worst match the stronger single-modality baseline.

For real spectroscopic data, distinguishing (a) from (b) requires demonstrating that a frozen pretrained spectral encoder outperforms one that has been jointly trained end-to-end on the same task. We report such experiments on FTIR and QCL tissue classification in Section 8. Our present study compares only two concrete mitigation strategies (freezing the spectral module, and Pezeshki's Spectral Decoupling Pezeshki et al. [2021]); broader comparisons against modality-specific learning rates, gradient surgery, GradNorm-style loss balancing, and related interventions are important future work.

4.6 Empirical support

Experiment 1.1 in Section 7 measures $\mathcal{D}_{\text{curv}}^{(M)}$ directly across a range of spatial widths and confirms monotonic growth with M: the spatial top eigenvalue grows $74\times$ over the sweep while the spectral top eigenvalue stays flat, driving the disparity into the thousands. The measured log–log slope of λ_ϕ against C_g is 0.70 ± 0.05, below the slope-1.0 prediction that applies to the tested architecture family (whose $C_g = \Theta(M)$); the deficit and its candidate causes are reported in Section 7. The qualitative prediction—unbounded spatial-block growth against a capped spectral block—is unambiguous in the data.

Remark 22 (Defense against linear triviality). A natural objection is that, since $f_{\boldsymbol{\theta}}$ is linear, the analysis must reduce to principal component analysis (PCA) of the input. This is not the case. PCA finds directions of maximum input variance *without reference to labels.* At initialization, however, the distinction is limited, and we state this candidly: the Gauss–Newton blocks carry no label information there (the softmax weights are near-uniform), and $G_{\boldsymbol{\theta}\boldsymbol{\theta}}$ is, by Lemma 18, an input-second-moment object propagated through a label-free random network. The defense against PCA-triviality is therefore a statement about *training*: along the trajectory the supervised residual shapes $\partial\hat{y}/\partial Z$ and hence the curvature, and the pathology we identify is precisely that this supervised signal fails to reach $\boldsymbol{\theta}$—not that $f_{\boldsymbol{\theta}}$ is incapable of representing useful features.

5 Theorem 2: Finite-Time Spectral Starvation

Theorem 14 is a geometric statement: as the spatial module widens, curvature concentrates in it and the spectral block's top curvature stays capped. This section states the dynamical consequence we actually observe in training—the spectral module barely moves during the period in which the training loss is being fit—as a theorem with *explicit dynamical hypotheses.* We deliberately do not claim that the geometry of Theorem 14 alone implies the dynamics: the bridge from curvature disparity to rate separation is stated as a hypothesis, motivated by Theorem 14 and verified empirically in Section 7. Deriving it from the architecture is left as an open problem (Section 9).

Two classical facts shape the formulation. First, for unregularized cross-entropy on separable data there is *no finite minimizer*: the loss decays like $1/t$ while the parameters diverge along the max-margin direction (Soudry et al. Soudry et al. [2018]). A theorem asserting convergence to a finite stationary point would therefore be vacuous or false in the regime modern networks inhabit. Second, the starvation phenomenon of Pezeshki et al. Pezeshki et al. [2021] is fundamentally about which features receive learning signal *during training*, not about the $t \to \infty$ limit. Both point the same way: the honest statement is a *finite-time* bound on how far the spectral parameters can move before the shortcut pathway has fitted the labels.

5.1 Setting and hypotheses

We analyze the continuous-time limit of joint training with a shared learning rate, in which the parameters $(\boldsymbol{\theta}_t, \boldsymbol{\phi}_t)$ evolve by gradient flow

$$\dot{\boldsymbol{\theta}}(t) = -\nabla_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}), \qquad \dot{\boldsymbol{\phi}}(t) = -\nabla_{\boldsymbol{\phi}}\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}). \tag{7}$$

All statements below hold for any absolutely continuous trajectory satisfying (7) for almost every t: under the smoothness of Assumption 3 the flow is classically well posed, and the almost-everywhere formulation additionally covers piecewise-smooth models such as ReLU networks, for which (7) is read as: at almost every t, $\dot{\boldsymbol{\theta}}(t) = -J_{\boldsymbol{\theta}}(t)^\top r(t)$ with $J_{\boldsymbol{\theta}}(t)$ a measurable selection of the limiting (Clarke) Jacobians. Every such selection obeys the velocity bound used in the proof: each limiting Jacobian is a limit of Jacobians attained within $\Phi_{\text{reg}} \times \mathcal{Z}_{\text{reg}}$ and hence inherits the operator-norm cap of Lemma 18. A discrete-SGD extension (Robbins–Monro tracking, with gradient noise entering the constants) is a natural further step in the standard stochastic-approximation framework (Borkar Borkar [2008], Ch. 2); we do not develop it here (Remark 48, supplement).

Recall from the chain rule (4) that $\nabla_{\boldsymbol{\theta}}\mathcal{L} = J_{\boldsymbol{\theta}}^\top r$ where r is the softmax residual, so

$$\left\|\dot{\boldsymbol{\theta}}(t)\right\| \;=\; \left\|J_{\boldsymbol{\theta}}(t)^\top r(t)\right\| \;\le\; \|J_{\boldsymbol{\theta}}(t)\|_{\text{op}} \cdot \|r(t)\|\,. \tag{8}$$

Fix a horizon $T > 0$ and suppose the trajectory remains in the regime of Assumption 7, i.e. $\boldsymbol{\phi}(t) \in \Phi_{\text{reg}}$ and $Z(t) \in \mathcal{Z}_{\text{reg}}$ for all $t \le T$ (*trajectory containment*). Under Assumptions 1, 4, and 7, Lemma 18 then caps the first factor over that horizon:

$$B_T \;:=\; \sup_{t \le T} \|J_{\boldsymbol{\theta}}(t)\|_{\text{op}} \;\le\; \sqrt{C_{\boldsymbol{\theta}}} \;=\; \widetilde{L}\,\sqrt{\lambda_{\max}(\Sigma_X)}, \tag{9}$$

a constant independent of the spatial width M, the parameter count C_g, and the horizon T. (Even without containment, $B_T < \infty$ for every finite T by continuity of $J_{\boldsymbol{\theta}}$ along the flow; containment is what makes the cap M*-independent*, which is the load-bearing property.) We write B for B_T when the horizon is clear. The remaining hypothesis concerns the second factor.

Remark 23 (Containment and the unregularized separable regime). Trajectory containment is a genuine restriction. In the unregularized separable regime that motivates the polynomial envelope below, parameter norms diverge—logarithmically for linear predictors (Soudry et al. Soudry et al. [2018]), and for homogeneous deep networks by the margin-maximization dynamics of Lyu and Li Lyu and Li [2020]—so a bound on $\|\partial g_{\boldsymbol{\phi}}/\partial Z\|_{\text{op}}$ that is uniform over *all* t cannot be assumed. Two standard settings restore it: weight decay or norm constraints, and the maximal-update parameterization Yang and Hu [2021], which leaves the flow unchanged (activation-level width-uniform stability is established there; the operator-norm form we need is a motivated assumption, per the discussion after Assumption 7). Weight decay and norm constraints, however, *modify* the flow (7) by an additional term the residual envelope does not control—under

L_2 decay on $\boldsymbol{\theta}$ the flow gains $-2\alpha\boldsymbol{\theta}$, and even at $r \equiv 0$ the displacement is $\boldsymbol{\theta}_0(1-e^{-2\alpha T}) \neq 0$—so Theorem 27 as stated applies only to the unregularized flow. A regularized variant would carry the extra displacement $\int_0^T \|2\alpha\, \boldsymbol{\theta}(t)\|\, dt$; equivalently, an experiment matching the theorem must place $\boldsymbol{\theta}$ in a zero-weight-decay parameter group, with decay (if any) on $\boldsymbol{\phi}$ only, which is what supplies containment. For homogeneous architectures within the scope of Lyu and Li Lyu and Li [2020], margin dynamics suggest $\widetilde{L}$ growing at most logarithmically in T, which would turn the displacement bound into $\mathcal{O}(\varepsilon \log^2(1/\delta))$; we do not claim this for attention-based $g_{\boldsymbol{\phi}}$, which is not homogeneous in the required sense.

Assumption 24 (Shortcut residual decay)**.** There exist constants $C \geq 1$ and $\mu > 0$ such that, along the joint trajectory (7) and on the horizon $[0, T]$ of any statement invoking this assumption,

$$\|r(t)\| \;\leq\; \frac{C}{1 + \mu t},$$

where $\|r(t)\|$ is the root-mean-square per-pixel residual of (3). We refer to μ as the *shortcut fitting rate*. The constants C and μ may depend on the initialization, the data, and the architecture—in particular μ is expected to grow with spatial capacity (Remark 25)—and are not assumed uniform over any family; the theorem holds for any trajectory admitting such an envelope.

Remark 25 (Why polynomial, and where μ comes from)*.* The $1/t$ envelope is the correct cross-entropy-compatible form: for logistic-type losses on separable data the training loss—and hence the residual norm—decays polynomially, not exponentially (Soudry et al. Soudry et al. [2018]). An exponential envelope, where valid (e.g. regularized or non-separable regimes), only strengthens the conclusion below. The hypothesis encodes the *shortcut* mechanism: the residual is driven down by the spatial module exploiting whatever signal is present in the feature map Z (positional and contextual structure), without requiring $f_{\boldsymbol{\theta}}$ to specialize—the spatial-signal sufficiency discussed in Section 5.4. The rate μ is expected to *grow* with spatial capacity—wider networks fit faster, the familiar regularity of over-parameterized training; kernel-regime analyses tie fitting rates to spectral properties of the model's kernel that improve with width (cf. the starvation analysis of Pezeshki et al. Pezeshki et al. [2021]), though we use no such derivation here. We test the capacity dependence of the fitting speed empirically in Experiment 1.2. Deriving $\mu(C_g)$ from the architecture is an open problem we state explicitly at the end of Section 5.4; the theorem takes μ as given.

Definition 26 (Fitting time)**.** For a target residual level $\delta \in (0, C)$, the fitting time is

$$T_\delta \;:=\; \inf\{\, t \geq 0 : \|r(t)\| \leq \delta \,\}.$$

Under Assumption 24, $T_\delta \leq (C/\delta - 1)/\mu$.

5.2 Main result

Theorem 27 (Finite-Time Spectral Starvation)**.** *Fix a horizon $T \geq 0$ and let $(\boldsymbol{\theta}(t), \boldsymbol{\phi}(t))$ be an absolutely continuous trajectory satisfying* (7) *for almost every $t \leq T$, contained in $\Phi_{\mathrm{reg}} \times \mathcal{Z}_{\mathrm{reg}}$ up to time T. Under Assumptions 1, 4, 7, and 24, with B_T as in* (9)*, the trajectory satisfies*

$$\|\boldsymbol{\theta}(T) - \boldsymbol{\theta}_0\| \;\leq\; \frac{B_T\, C}{\mu}\, \log(1 + \mu T),$$

and, whenever $T \geq T_\delta$, at the fitting time T_δ,

$$\|\boldsymbol{\theta}(T_\delta) - \boldsymbol{\theta}_0\| \;\leq\; \frac{B_T\, C}{\mu}\, \log\!\left(\frac{C}{\delta}\right).$$

In particular, writing $\varepsilon := B_T/\mu$ for the effective module-rate ratio, the spectral displacement by the time the training residual reaches level δ is $\mathcal{O}(\varepsilon \log(1/\delta))$.

Proof. By (8), the operator-norm cap (9), and Assumption 24, for almost every $t \le T$,

$$\left\|\dot{\boldsymbol{\theta}}(t)\right\| \;\le\; B_T \cdot \|r(t)\| \;\le\; \frac{B_T\, C}{1+\mu t}.$$

Integrating,

$$\|\boldsymbol{\theta}(T) - \boldsymbol{\theta}_0\| \;\le\; \int_0^T \left\|\dot{\boldsymbol{\theta}}(t)\right\| dt \;\le\; B_T\, C \int_0^T \frac{dt}{1+\mu t} \;=\; \frac{B_T\, C}{\mu} \log(1+\mu T).$$

For the second claim, Definition 26 gives $1 + \mu T_\delta \le C/\delta$, so $\log(1+\mu T_\delta) \le \log(C/\delta)$. □

Corollary 28 (Attribution of the loss decrease)**.** *Under the hypotheses of Theorem 27, with $g_{\boldsymbol{\phi}}$ smooth jointly in $(\boldsymbol{\phi}, Z)$ (the smooth branch of Assumption 3) so that $t \mapsto \mathcal{L}(t)$ is differentiable along the joint flow, and whenever $\mathcal{L}(T) < \mathcal{L}(0)$, the energy identity $-\frac{d}{dt}\mathcal{L} = \|\nabla_{\boldsymbol{\theta}}\mathcal{L}\|^2 + \|\nabla_{\boldsymbol{\phi}}\mathcal{L}\|^2$ holds, and the spectral module's cumulative share of the loss decrease obeys*

$$\mathcal{A}_{\boldsymbol{\theta}}(T) \;:=\; \frac{\int_0^T \|\nabla_{\boldsymbol{\theta}}\mathcal{L}(t)\|^2\, dt}{\mathcal{L}(0) - \mathcal{L}(T)} \;\le\; \frac{B_T^2\, C^2}{\mu\left(\mathcal{L}(0) - \mathcal{L}(T)\right)}.$$

Proof. By (8) and Assumption 24, $\|\nabla_{\boldsymbol{\theta}}\mathcal{L}(t)\|^2 \le B_T^2 C^2/(1+\mu t)^2$ for almost every $t \le T$. Integrating, $\int_0^T \|\nabla_{\boldsymbol{\theta}}\mathcal{L}\|^2\, dt \le B_T^2 C^2 \int_0^\infty (1+\mu t)^{-2}\, dt = B_T^2 C^2/\mu$, and the energy identity makes the denominator the total dissipation, of which the numerator is the spectral part. □

Remark 29 (Two instantiations, one honest caveat)*.* Corollary 28 is the asymmetry statement Theorem 27 alone does not provide: the displacement bound has the same *form* for both modules, whereas the attribution bound's constant is capacity-independent only for $\boldsymbol{\theta}$. It admits two instantiations. With the a priori cap $B_T \le \widetilde{L}\sqrt{\lambda_{\max}(\Sigma_X)}$ of Lemma 18 the bound may exceed 1 and thus be vacuous at realistic constants—we report the ratio rather than hide it. With the *realized* gain $\widehat{B}_T := \sup_{t \le T} \|\nabla_{\boldsymbol{\theta}}\mathcal{L}(t)\| \,/\, \|r(t)\| \le B_T$, measurable from any training run, it becomes a per-run diagnostic. Both are computable from a run's logs.

5.3 Interpretation

Theorem 27 is a statement about *training-time* behavior, which is what practice cares about: training runs end when the loss is fit (or by early stopping), not at $t = \infty$.

The starvation regime. The displacement bound is a product of factors with opposite capacity dependence. The factor B_T is capped independently of C_g (Lemma 18): no matter how large the spatial module, the spectral gradient cannot exceed $B_T\, \|r\|$. The fitting rate μ is *expected* to grow with spatial capacity (Remark 25; this is an empirical regularity we test, not a theorem): wider shortcut pathways drive the residual down faster. For families of architectures along which the envelope amplitude C stays bounded—as it does empirically, since $\|r(0)\|$ is $\Theta(1)$ at near-uniform initial predictions and the observed residuals decay from the start rather than plateauing—the displacement at the fitting time scales as $\varepsilon \log(1/\delta)$ with $\varepsilon = B_T/\mu \to 0$, and the spectral module is increasingly pinned to its initialization *for the entire useful duration of training.* We state the boundedness of C along the family as an explicit condition rather than a consequence: Assumption 24 deliberately does not make its constants uniform over architectures, so the capacity-asymptotic reading requires it. This is the capacity-worsens-starvation prediction tested in Experiment 1.2.

What we do not claim. We make no claim about the $t \to \infty$ limit. For unregularized cross-entropy on separable data no finite stationary point exists (Soudry et al. Soudry et al. [2018]), and $\boldsymbol{\theta}$ may eventually move substantially over horizons exponentially longer than any practical training run ($T = \mathcal{O}(\varepsilon^{-q})$ still yields displacement $\mathcal{O}(\varepsilon \log(1/\varepsilon)) \to 0$, but larger horizons are unconstrained). The finite-time framing is not a weakness: it is the phenomenon. A spectral module that receives its learning signal only after the loss has been fitted receives it too late to shape which features the classifier uses.

The logarithm is benign. For any polynomial training horizon $T = \mathcal{O}(\varepsilon^{-q})$ with fixed q, and along families with bounded envelope amplitude C as above, the displacement is $\mathcal{O}(\varepsilon \log(1/\varepsilon))$, which vanishes as $\varepsilon \to 0$. The bound is thus meaningful across every practically reachable horizon, not merely at a single stopping time.

Relation to lazy training. Structurally, Theorem 27 is a *module-wise lazy training* bound: a bounded module Jacobian multiplied by an integrable residual envelope yields logarithmic parameter movement, in the spirit of Chizat, Oyallon, and Bach Chizat et al. [2019]. The differences are the driver and the scope. In the lazy-training literature, small movement follows from a scaling of the *model* (width or output scale) that makes the linearization accurate for *all* parameters; here, the bound is *informative* for one module only. The identical integration applied to $\boldsymbol{\phi}$ carries $\sup_t \|J_{\boldsymbol{\phi}}(t)\|_{\mathrm{op}}$ in place of the cap, so the asymmetry lies in the constants—the spectral factor is capacity-independent (Lemma 18) while the spatial factor is free to grow with capacity—not in the form of the inequality. Whether the constants actually separate at a given scale is an empirical question (Section 7). The phenomenon, where present, is a *selective* laziness induced by the compositional position of the spectral module, not a global linearization regime.

5.4 On the hypotheses

The operator-norm cap is proven, not assumed. The factor B_T is supplied by Lemma 18 under Assumptions 1 and 7. We are precise about how much of Section 4 enters this proof: only the cap—the same lemma that bounds $\mathcal{D}_{\mathrm{curv}}$'s denominator. The width-linear growth of the spatial block (Lemma 16) appears nowhere in the derivation; its role is to motivate the capacity dependence of μ, which remains a hypothesis. Theorem 27 is a consequence of the cap plus the envelope, not of the disparity.

Residual decay encodes the shortcut. Assumption 24 is a statement about the *residual only*: the training loss gets fitted at rate μ along the joint trajectory. It does not mention $\boldsymbol{\theta}$, and in particular it does not assume the spectral module stays put. The shortcut interpretation—that the fitting is done by the spatial module exploiting structure already present in Z (positional cues, contextual patterns)—is the mechanism that makes the assumption plausible at high spatial capacity, and it is satisfied in practice whenever the spatial model can overfit the training set, the norm for modern over-parameterized CNNs and ViTs. We test the envelope itself empirically in Experiment 1.2.

Remark 30 (Is the hypothesis circular?). A natural worry is that assuming fast residual decay comes close to assuming the conclusion. It does not, and the direction of the implication is the theorem's content. The assumption constrains what the *loss* does; the theorem then caps how far $\boldsymbol{\theta}$ can have travelled *on any trajectory consistent with that loss behavior*—including trajectories on which $\boldsymbol{\theta}$ itself does part of the fitting. Nothing in the hypothesis privileges the spatial module: if a trajectory fits the residual at rate μ partly through spectral learning, the bound still applies and shows the total spectral displacement incurred in doing so is at most $(B_T C/\mu)\log(C/\delta)$. Read contrapositively, this is the sharpest form of the claim: *substantial* *spectral learning by the fitting time requires slow fitting.* A trajectory on which $\boldsymbol{\theta}$ moves far by T_δ must have small μ—the loss must stay unfit for a long time—which is precisely what freezing-based pipelines engineer and what fast-fitting joint training precludes. (Over horizons far beyond the fitting time the logarithm can grow without bound; see "What we do not claim" below.) The hypothesis is also independently checkable: the envelope is a measurable property of a training run, verified or refuted by the run's own residual curve, with no reference to $\boldsymbol{\theta}$.

Proposition 31 (The witness direction is a residual-fitting direction)**.** *Suppose g_ϕ ends in a dense classifier head over penultimate features $h_n \in \mathbb{R}^{M_h}$ with $M_h = \Theta(M)$, as in Lemma 16. In the normalized logit coordinates $\tilde{y} := N^{-1/2}(\hat{y}_1, \ldots, \hat{y}_N)$ of* (3)*, gradient flow moves the logits by $\dot{\tilde{y}} = -\widehat{\Theta}\, r$, with $\widehat{\Theta} := JJ^\top$ the normalized empirical tangent kernel. Let $c \perp \mathbf{1}$ be a unit class contrast and $w := N^{-1/2}(c, \ldots, c)$ the unit direction applying c at every pixel (the* class-prior direction*). If the mean penultimate feature $\bar{h} := N^{-1}\sum_n h_n$ satisfies $\left\|\bar{h}\right\|^2 \geq \bar{q}\, M_h$ at initialization—as holds with probability $1 - \mathcal{O}(1/M_h)$ for the instance of Proposition 38 (Step A, with $\bar{q} = \sigma_b^2/(4\pi)$)—then*

$$w^\top \widehat{\Theta}\, w \;=\; \left\|J^\top w\right\|^2 \;\geq\; \left\|\bar{h}\right\|^2 \;\geq\; \bar{q}\, M_h \;=\; \Omega(M):$$

the tangent-kernel quadratic form along the class-prior direction grows linearly in the spatial width.

Proof. The chain rule with (7) gives $\dot{\tilde{y}} = N^{-1/2}(\partial \hat{Y}/\partial p)\, \dot{p} = J \cdot (-J^\top r)$, the first display. For the second, restricting $\left\|J^\top w\right\|^2$ to the head-weight coordinates only lowers it. With $\hat{y}_n = W h_n + b$ and $\partial \hat{y}_{n,i}/\partial W_{i'j} = \delta_{ii'} h_{n,j}$, the head-weight block of $J^\top w$ is $N^{-1}\sum_n c\, h_n^\top = c\, \bar{h}^\top$, whose squared Frobenius norm is $\|c\|^2 \left\|\bar{h}\right\|^2 = \left\|\bar{h}\right\|^2$. □

Remark 32 (What the proposition does and does not add)*.* Proposition 31 is a first link between Theorem 14's geometry and the *content* of Assumption 24, and we state its limits plainly. It is an instantaneous, initialization-time statement: it identifies the class-prior contrast—the constant-across-pixels component of the residual, i.e. the majority-class shortcut—as a direction the wide spatial module is geometrically fastest to fit, at a rate growing linearly in width. It does not control the trajectory, does not derive the envelope of Assumption 24, and concerns one component of r: the evolution $\dot{r} = H_\tau \dot{\tilde{y}}$ couples all components, so the decay of $\langle w, r\rangle$ along the flow also depends on the rest of the residual. What it converts is the reading of Theorem 14: the width-linear curvature direction is not merely capacity to move (Remark 40) but capacity to move *along a residual-fitting direction*—exactly the mechanism Assumption 24 postulates and Experiment 1.2 measures.

Rate separation is motivated, not derived. Theorem 14 shows curvature disparity grows with capacity; it does not by itself prove that μ grows or that the trajectory satisfies Assumption 24. We keep the two theorems logically independent: Theorem 14 identifies a capacity-dependent geometric mechanism *capable* of generating disparate module learning rates; Theorem 27 characterizes the starvation dynamics *when* such a separation is present. Experiments 1.1–1.3 test the mechanism and the dynamics separately. Deriving Assumption 24—and in particular the growth of μ with C_g—from the architecture (e.g. via an NTK analysis of the spatial pathway at large width, adapting Pezeshki et al. Pezeshki et al. [2021]) is, in our view, the most valuable missing theorem in this program: it would chain Theorems 14 and 27 into a single causal statement. We state it as an open problem rather than sketch it (Remark 48, supplement), and we test its content empirically: Experiment 1.2 measures the fitting speed as a function of spatial capacity.

6 The Effective Gradient Ratio Diagnostic

Theorems 14 and 27 together identify a training pathology that is invisible to *single-run* loss-curve monitoring. The failure is counterfactual: a jointly trained model converges to an apparently healthy solution—training loss down, validation metrics respectable, every curve shaped like a normal run—that is nonetheless worse than what the *same architecture* achieves with its spectral module frozen. Nothing in the run's own curves flags this, because the deficit is visible only against a baseline the practitioner has not trained. One discovers it by training the frozen counterpart (twice the compute, and only if one thinks to try), or not at all.

This section introduces the *Effective Gradient Ratio* (EGR), a real-time diagnostic that detects gradient starvation as it occurs. EGR is a single scalar, computable inside any training loop at negligible cost, that converts the finite-time starvation prediction of Theorem 27 into a concrete observable.

6.1 Definition and basic properties

Recall from Definition 10 that at training step t,

$$\mathrm{EGR}(t) = \frac{\|\nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}_t, \boldsymbol{\phi}_t)\|}{\|\nabla_{\boldsymbol{\phi}} \mathcal{L}(\boldsymbol{\theta}_t, \boldsymbol{\phi}_t)\|}.$$

The numerator and denominator are computable directly from gradients accumulated during the standard backward pass. EGR is unitless and invariant to the choice of learning rate.

Initial value: a heuristic. A useful rule of thumb for the starting point of the diagnostic is

$$\mathrm{EGR}(0) \;\sim\; \sqrt{C_f / C_g} \qquad \text{(heuristic, not a theorem).} \tag{10}$$

The reasoning is a parameter-counting one: at variance-scaled initialization each module's gradient has entries of comparable typical magnitude, so the squared gradient norms scale with the number of parameters carrying them, $\|\nabla_{\boldsymbol{\phi}} \mathcal{L}\|^2 \sim C_g$ against $\|\nabla_{\boldsymbol{\theta}} \mathcal{L}\|^2 \sim C_f$, and the square root comes from passing to norms. We state (10) as a heuristic rather than a proposition for three reasons: it exchanges the expectation of a ratio for the ratio of expectations without a concentration argument; the per-entry-magnitude premise is parametrization-dependent and would change under μP, which we cite elsewhere Yang and Hu [2021]; and the spectral module's gradient additionally passes through the bottleneck, which the count ignores. Its role in this paper is limited to setting the scale for the *relative* threshold of Definition 36, which compares $\mathrm{EGR}(t)$ against the measured $\mathrm{EGR}(0)$ of the run at hand and so does not depend on (10) being sharp. The corresponding numerical value for our motivating architecture follows from the capacity ratio reported in Section 8.

6.2 Decay of the spectral gradient

Proposition 33 (Spectral gradient decay)**.** *Under the hypotheses of Theorem 27—Assumptions 1, 4, 7, and 24, with the trajectory contained in $\Phi_{\mathrm{reg}} \times \mathcal{Z}_{\mathrm{reg}}$ up to a horizon T—we have, for every $t \le T$,*

$$\|\nabla_{\boldsymbol{\theta}} \mathcal{L}(t)\| \;\le\; \frac{B_T\, C}{1 + \mu t}, \qquad B_T \;\le\; \sqrt{C_{\boldsymbol{\theta}}} \;=\; \widetilde{L}\sqrt{\lambda_{\max}(\Sigma_X)},$$

with B_T as in (9)*, $C_{\boldsymbol{\theta}}$ the cap of Lemma 18, and C, μ the constants of Assumption 24.*

Proof. Identical to (8): $\|\nabla_{\boldsymbol{\theta}} \mathcal{L}\| = \left\|J_{\boldsymbol{\theta}}^\top r\right\| \le \|J_{\boldsymbol{\theta}}\|_{\mathrm{op}} \|r\| \le B_T \|r\|$ for $t \le T$, and Assumption 24 bounds $\|r(t)\|$. $\square$

Remark 34. The horizon restriction is not a technicality inherited for tidiness: Remark 23 notes that in the unregularized separable regime—the regime Assumption 24 models—a cap uniform over all $t \in [0, \infty)$ cannot be assumed. In practice T is the training run's length, which is exactly the horizon the diagnostic operates over.

Remark 35 (From gradient decay to the EGR ratio). Proposition 33 bounds the *numerator* of the EGR. The denominator $\|\nabla_{\boldsymbol{\phi}}\mathcal{L}\| = \left\|J_{\boldsymbol{\phi}}^{\top} r\right\|$ is also proportional to the residual, so both quantities vanish together as the loss is fitted, and the behavior of their ratio is not determined by residual decay alone: it depends on the relative conditioning of the two Jacobians along the trajectory. We therefore treat the EGR trajectory as an empirical object, characterized in Section 6.5, and use Proposition 33 only for the absolute claim that the spectral module's learning signal decays polynomially while the loss is being fitted.

6.3 Threshold criterion

The absolute decay of the spectral gradient (Proposition 33), combined with the empirical behavior of the ratio characterized in Section 6.5, motivates a practical criterion for detecting the onset of gradient starvation. We emphasize its status: the threshold rule is an empirically calibrated heuristic that the theory motivates but does not derive (Remark 35).

Definition 36 (Gradient starvation alert)**.** We say the model is in a *gradient starvation regime* at step t if

$$\mathrm{EGR}(t) \quad < \quad \tau \cdot \mathrm{EGR}(0),$$

where $\tau \in (0, 1)$ is a user-supplied tolerance. Empirically, we recommend $\tau \in [0.05, 0.1]$, corresponding to a 10–20× drop in EGR relative to initialization.

Remark 37. The threshold τ is the only hyperparameter of the diagnostic. The default $\tau = 0.1$ corresponds to a one-decade decay and is the value we use throughout Section 7.

Practical use. A practitioner logs $\mathrm{EGR}(t)$ at each step. When the alert fires, two options are available:

1. **Freeze the spectral module.** This is the prescription most directly motivated by Theorem 27: removing $\boldsymbol{\theta}$ from the optimization eliminates the timescale gap. We discuss this in Section 9.
2. **Inject a fixed residual baseline.** A softer intervention: replace $Z = f_{\boldsymbol{\theta}}(X)$ with $Z = f_{\boldsymbol{\theta}}(X) + h(X)$ for a fixed feature extractor h (e.g. PCA). The fixed baseline ensures the spatial model receives stable input even if $f_{\boldsymbol{\theta}}$ continues to drift, breaking the shortcut without freezing.

We evaluate both interventions in Experiment 1.3.

6.4 Relationship to existing diagnostics

EGR is conceptually related to several existing notions but is distinct in its scope and intent.

GradNorm Chen et al. [2018]. GradNorm dynamically reweights losses in multi-task learning to equalize gradient magnitudes across tasks. EGR measures rather than intervenes, and applies to single-task compositional architectures rather than multi-task losses.

Modality-specific learning rates (MSLR). In multimodal learning, MSLR adjusts learning rates per modality. EGR diagnoses the underlying pathology that MSLR addresses but in a unimodal, two-stage architecture where the "modalities" are the spectral and spatial subspaces of a single input.

Loss curve monitoring. The conventional practice of monitoring training and validation loss fails to detect gradient starvation because both can decay even as $\boldsymbol{\theta}$ stagnates. EGR is sensitive to the structural failure even when no global loss curve abnormality is present.

6.5 Empirical validation: EGR as a diagnostic

We validate EGR as a diagnostic of training pathology in two stages. First, we establish that EGR statistics correlate with generalisation across our full sweep. Second, we test whether the signal is genuine or merely a re-encoding of model capacity.

Pooled correlation (capacity confounded). Across $N = 24$ runs spanning four widths $\{16, 64, 256, 1024\}$ with six seeds each, EGR depth correlates strongly with final test accuracy: Pearson $r = -0.723$ ($p = 6.6 \times 10^{-5}$), Spearman $\rho = -0.808$ ($p = 1.8 \times 10^{-6}$), and with the train–test gap at $r = +0.741$ ($p = 3.5 \times 10^{-5}$). However, EGR depth is itself strongly correlated with $\log$ width ($r = +0.903$): roughly 77% of the pooled correlation with test accuracy is attributable to capacity rather than to EGR depth per se (partial r controlling for $\log$ width collapses to -0.169, $p = 0.44$). The pooled correlation alone cannot establish a capacity-independent diagnostic.

Fixed-capacity test. To isolate the capacity-independent signal, we ran a second experiment holding CNN width fixed at $D = 256$ and varying data noise $\{0.05, 0.10, 0.15, 0.20\}$, learning rate $\{3{\times}10^{-4}, 10^{-3}, 3{\times}10^{-3}\}$, weight decay $\{0, 10^{-3}\}$, and six seeds, for a total of $N = 144$ runs. At fixed capacity, $\mathrm{EGR}_{\mathrm{min}}$ is the relevant statistic (with width fixed, egr_depth reduces to a constant offset of egr_min in expectation). It exhibits a within-capacity correlation with generalisation that is statistically robust but practically moderate: Pearson $r(\mathrm{EGR}_{\mathrm{min}}, \mathrm{acc}) = +0.344$ ($p = 2.5 \times 10^{-5}$).

The relationship survives every adversarial control we applied. $\mathrm{EGR}_{\mathrm{min}}$ is essentially orthogonal to the swept hyperparameters — marginal correlations with noise, learning rate, and weight decay are $+0.029$, -0.073, and $+0.007$ respectively, with the full hyperparameter set explaining only $R^2 = 0.015$ of $\mathrm{EGR}_{\mathrm{min}}$ variance. The partial correlation $r(\mathrm{EGR}_{\mathrm{min}}, \mathrm{acc} \mid \mathrm{noise}, \mathrm{lr}, \mathrm{wd}) = +0.395$ ($p = 9.5 \times 10^{-7}$) is *larger* than the pooled $r = +0.334$, which is the opposite signature of a hidden hyperparameter confound. The within-bin analysis is the strongest evidence: across the 24 $(\mathrm{noise}, \mathrm{lr}, \mathrm{wd})$ cells in which only the seed varies, 23 of 24 yield strongly positive $r(\mathrm{EGR}_{\mathrm{min}}, \mathrm{acc})$ (mean $+0.584$, Fisher-z aggregated $t = 14.65$, $p < 10^{-13}$).

Controlling for final training loss — itself a non-trivial predictor with $r = +0.47$ for accuracy and $r = -0.55$ for the overfit gap — leaves the $\mathrm{EGR}_{\mathrm{min}}$ partial correlation essentially unchanged at $+0.335$ ($p = 4.1 \times 10^{-5}$). EGR therefore carries information that train-loss monitoring does not.

Practical interpretation. Honesty about effect size is required. Treated as a binary alarm, $\mathrm{EGR}_{\mathrm{min}}$ achieves ROC-AUC $= 0.612$ for flagging below-median test accuracy, and $\mathrm{EGR}_{\mathrm{depth}}$ achieves only 0.538 for high overfit gap. The best operating point ($\mathrm{EGR}_{\mathrm{min}} < 0.10$) catches 90% of bad runs but also flags 57% of healthy runs — no threshold delivers a usable trade-off as a standalone indicator.

Verdict. At fixed capacity, EGR is a real, capacity-independent signal correlated with generalisation. The within-capacity effect size is small enough that EGR is best deployed alongside train-loss monitoring as a complementary diagnostic, not as a standalone run-quality indicator. For research and ablation work, EGR provides genuine insight into the gradient starvation mechanism beyond what loss monitoring reveals; for production monitoring of single training runs, plain loss curves remain the primary signal and EGR a useful adjunct.

6.6 Implementation

The EGR diagnostic is implemented as a PyTorch training callback. It hooks into the standard training loop between the backward pass and the optimizer step:

```
loss.backward()
egr_logger.log_step(global_step)   # <-- the only added line
optimizer.step()
optimizer.zero_grad()
```

The callback identifies the spectral and spatial parameter groups via `model.spectral.parameters()` and `model.spatial.parameters()` (or any user-supplied naming convention) and records the gradient norms and EGR at each step. We release the implementation as the `egr` package, available at the project repository. [TODO: Phase 4, W12: prepare PyPI package skeleton. Add unit tests for gradient norm computation and threshold detection.]

7 Synthetic Experiments

We validate Theorems 14 and 27 and the EGR diagnostic on a controlled synthetic spectral-spatial classification problem. Three experiments isolate distinct theoretical predictions: (1.1) the Hessian condition number bound; (1.2) the two-timescale dynamics under joint training; (1.3) the equal-information killer test demonstrating that joint training prefers spatial features even when spectral and spatial signals carry equal information.

7.1 Synthetic data and architectures

We use a calibrated generator of synthetic hyperspectral images $X \in \mathbb{R}^{H\times W\times S}$ with $H = W = 16$ and $S = 64$, supporting two-class per-pixel classification with controlled spectral and spatial signal strengths through scalar parameters α and β. The data generator places a known spectral direction $u \in \mathbb{R}^S$ in each pixel scaled by a per-pixel latent $z \sim \mathcal{N}(0, 1)$ and modulated by α, with smooth per-class spatial templates modulated by β entering the label-generating logits. Bayes-optimal accuracy is ≈ 0.85–0.95 at $\alpha = \beta = 1$ and noise level 0.1.

For the spatial module $g_{\boldsymbol{\phi}}$ we use three architectures: a two-layer convolutional network (SpatialCNN, 3×3 kernels), a two-layer pre-LN Vision Transformer (SpatialViT, pixel-token), and a per-pixel MLP baseline (SpatialMLP, no spatial receptive field). The MLP is included for ablation; it cannot exhibit a spatial shortcut by construction.

7.2 Experiment 1.1: Hessian capacity ablation

We vary the spatial width D of a SpatialMLP over nearly three orders of magnitude ($D \in \{16, 32, 64, \ldots, 8192\}$, a $512\times$ range) with $C_f = 1024$ held fixed; all measurements in this experiment use the SpatialMLP family. For each (D, seed) we compute the top eigenvalue of the spectral and spatial diagonal blocks of the loss Hessian via power iteration (5 seeds per width). Theorem 14 is stated for the Gauss–Newton blocks; the full-Hessian blocks additionally contain a residual-weighted functional term, and we flag this measurement gap explicitly. The predictions to be tested are $\lambda_{\max}(J_{\boldsymbol{\phi}}^\top J_{\boldsymbol{\phi}}) = \Omega(M)$ (Lemma 16) and $\lambda_{\max}(J_{\boldsymbol{\theta}}^\top J_{\boldsymbol{\theta}})$ approximately constant (Lemma 18).

Findings. Across 50 measurements:

- $\lambda_{\max}(J_{\boldsymbol{\theta}}^\top J_{\boldsymbol{\theta}})$ is flat at ≈ 0.02 over the full range of D – the spectral block does not grow with the spatial capacity, consistent with Lemma 18.
- $\lambda_{\max}(J_{\boldsymbol{\phi}}^\top J_{\boldsymbol{\phi}})$ grows by $74\times$ (from 0.62 to 46) as C_g grows by $510\times$.

- The empirical log-log slope of $\lambda_{\max}(J_\phi^\top J_\phi)$ vs. C_g is 0.70 ± 0.05 across the full range. For the SpatialMLP family $C_g = \Theta(D)$ (input and output dimensions are fixed), so the width-linear law of Lemma 16 predicts slope 1.0 against C_g; the measured 0.70 is sublinear, short of the width-linear asymptote. Candidate causes are finite-width corrections, the loss-Hessian versus Gauss–Newton measurement gap flagged above, and the correlated non-Gaussian features Z relative to the mean-field idealization. The $\sqrt{C_g}$ (slope-0.5) form of Theorem 14 applies only to fixed-depth $\Theta(M^2)$ families, which this sweep does not include.

- The module curvature disparity $\mathcal{D}_{\text{curv}} = \lambda_\phi/\lambda_\theta$ grows from 40 at $C_g/C_f = 0.3$ to 2,407 at $C_g/C_f = 152$. At the capacity ratios of real spectroscopy pipelines (Section 8), this extrapolates to disparities in the thousands.

The qualitative geometric prediction is verified: the spectral and spatial block eigenvalues diverge with capacity, producing a severe module curvature disparity. The measured scaling exponent is sublinear relative to the width-linear asymptote; the qualitative divergence, not the exact exponent, is what Theorems 14 and 27 require, and we report the quantitative deficit as an open discrepancy.

Headline figure. Figure **??** (paired view: λ_ϕ and λ_θ on the same axes across the D sweep) shows the two eigenvalues diverging as D grows. The constancy of λ_θ is itself a strong qualitative finding: the spectral block carries no growing curvature with D, isolating the asymmetry to the spatial side.

7.3 Experiment 1.2: Two-timescale training dynamics

We train the compositional model under joint and frozen-spectral conditions across multiple capacities and architectures, logging the EGR every step. The predictions to be tested are Proposition 33 (the spectral gradient norm decays as $C/(1+\mu t)$ while the loss is fitted) and the consequence that the EGR collapse depth should scale monotonically with capacity.

Findings — CNN. On SpatialCNN at widths $D \in \{16, 64, 256, 1024\}$ with 3 seeds each:

- EGR collapse depth is monotonic in D:

$$\text{EGR}_{D=16}^{\text{final}} \approx 0.40, \quad \text{EGR}_{D=64}^{\text{final}} \approx 0.25, \quad \text{EGR}_{D=256}^{\text{final}} \approx 0.10, \quad \text{EGR}_{D=1024}^{\text{final}} \approx 0.08.$$

- Joint training final test loss diverges at high capacity. At $D = 1024$, joint test loss $= 2.27$ while frozen test loss $= 1.15$.

- Joint test accuracy peaks early (≈ 0.70) then decays as the model overfits to shortcuts. Frozen test accuracy is more stable.

This is the cleanest empirical validation of Theorem 27: on the canonical CNN architecture, the EGR exhibits exactly the capacity-monotonic collapse anticipated by Proposition 33 and the starvation regime of Theorem 27.

Findings – ViT. On SpatialViT, training enters a full memorization regime: train loss reaches 0.01–0.03 at all widths, while test loss is catastrophic (2.5–5.0). In this regime the EGR observable loses interpretive value: both gradient norms approach the noise floor and their ratio is dominated by stochasticity. The underlying prescription, however, still holds: frozen variants beat joint variants at every width tested.

Conclusion. Theorem 27's qualitative prediction (joint training fails by undertraining the spectral module) is universal: it manifests on both CNN and ViT spatial models. The specific EGR observable cleanly tracks the predicted decay in non-memorization regimes (CNN at our capacities) and becomes uninformative in the memorization regime (ViT). We discuss this limitation in Section 9.

7.4 Experiment 1.3: The equal-information killer test

The previous experiments validate the geometric (Theorem 14) and dynamical (Theorem 27) predictions on capacity-asymmetric compositional models. Experiment 1.3 tests a stronger qualitative claim suggested by the theory: *when spectral and spatial signals carry equal information about the label, end-to-end joint training will underperform freezing because of the shortcut dynamics.*

Calibration. We require an operational definition of "equal information" between the spectral pathway (per-pixel content of X along a direction $u \in \mathbb{R}^S$) and the spatial pathway (per-pixel content of X along an orthogonal direction $v \perp u$, with strength modulated by a smooth per-position function of the class). We compare three modes:

- *bayes*: scalar strengths α, β chosen such that a single-modality logistic regression on α-only data and a position-majority classifier on β-only data both achieve a target accuracy (0.75 in our experiments);
- *ntk*: α, β chosen such that the leading Gram-matrix eigenvalues of the per-pixel features from the two modalities are equal;
- *margin*: α, β chosen such that the L2 margins of single-modality logistic regressions are equal.

Setup. For each calibration mode we train the canonical CNN composition at width $D = 256$ for 150 epochs on $N_{\text{train}} = 512$ samples, repeated over 3 seeds. The four training conditions per mode are:

- *joint*: end-to-end training with both modules trainable;
- *frozen*: spectral module frozen at random initialization;
- *spectral-only*: $\beta = 0$, only spectral signal in X;
- *spatial-only*: $\alpha = 0$, only spatial signal in X.

Results. Under the `bayes` calibration we find:

- The single-modality baselines reach equal accuracy by construction: $\alpha = 0.82$, $\beta = 18.75$ produces spectral-only $= 0.746$, spatial-only $= 0.754$ (logistic regression and position-majority bounds respectively).
- Empirically on the full CNN model: spectral-only training (data with $\beta = 0$) achieves test accuracy 0.53, spatial-only training (data with $\alpha = 0$) achieves test accuracy 0.53. The CNN-achievable accuracies are equal, confirming the calibration is meaningful in architecture.
- Frozen-spectral training on the full data achieves 0.61, exploiting both modalities.
- **Joint training drops to** 0.48 – below both single-modality baselines (0.53) and 0.13 points below the frozen variant.

Two honest caveats bound this result. First, the effect size is modest in absolute terms: with 3 seeds and 128 test samples the per-condition standard error is roughly 0.02, so the 0.05 gap to the single-modality baselines is resolvable but not dramatic, and 0.48 is statistically indistinguishable from the 0.50 chance floor. Second, the CNN reaches only 0.53 on single-modality data against a 0.75 calibration target, so the network operates far from the Bayes bound and part of the joint-training deficit could in principle reflect generic overfitting with more trainable parameters. What the direction of the result nevertheless establishes is that joint training does not merely fail to combine the two modalities: it undershoots baselines that demonstrably use only one, which is consistent with active corruption of the spectral pathway rather than mere under-use, and is inconsistent with the "spectral features are simply irrelevant" reading (Remark 21). The frozen-vs-joint ordering, which is the theory's core prescription, holds in all three calibration modes (Table 1: +0.13, +0.04, +0.09); the undershoot below both single-modality baselines appears specifically in the `bayes` mode, where the modalities are matched most cleanly.

Comparison of calibration modes. Table 1 compares the three calibration modes on two criteria: (i) calibration quality, defined as $1 - |a_{\mathrm{spec}} - a_{\mathrm{spat}}|$ where $a_{\mathrm{spec}}, a_{\mathrm{spat}}$ are the achieved single-modality accuracies; and (ii) shortcut strength, defined as $a_{\mathrm{frozen}} - a_{\mathrm{joint}}$.

mode	α, β	spec / spat	joint	frozen	shortcut
bayes	0.82, 18.75	0.75/0.75	0.48	0.61	**+0.13**
ntk	0.99, 4.91	0.78/0.58	0.53	0.57	+0.04
margin	5.00, 5.00	0.95/0.59	0.60	0.69	+0.09

Table 1: Calibration mode comparison for the equal-information killer test (CNN $D = 256$, 3 seeds). *spec / spat*: achievable single-modality accuracies. *shortcut* $:= a_{\mathrm{frozen}} - a_{\mathrm{joint}}$. The `bayes` mode produces both the cleanest modality equalization and the largest shortcut effect.

Conclusion. The `bayes` calibration delivers the qualitative claim: when spectral and spatial signals carry equal information about the label (measured by what each modality alone can provide), end-to-end joint training underperforms both single-modality baselines, while freezing the spectral module recovers information that joint training squanders. Subject to the effect-size caveats above, this is empirical support consistent with the spectral-shortcut mechanism suggested by Theorem 27.

7.5 Robustness

In supplementary experiments we vary the optimizer (Adam vs. SGD with momentum), the noise level (0.05 vs. 0.10), and the spectral input dimension S. The capacity-driven joint-vs-frozen gap and the EGR collapse pattern hold across all variations, with two notable modulations: SGD makes the EGR signal cleaner (Adam normalizes per-parameter and partially masks the natural gradient ratio), and lower noise allows cross-entropy to fully saturate, accelerating the emergence of the shortcut.

[TODO: Phase 5, W15: write the robustness analyses into the supplement once Exp 1.3 is finalized.]

8 Real-Data Study: Geometry, Dynamics, and Scope

The synthetic experiments in Section 7 verify the theorems where their assumptions hold. This section takes the theory to a production-scale spectral-spatial pipeline: per-pixel 4-class tissue classification on QCL infrared hyperspectral images of breast cancer tissue microarrays. It contains two kinds of evidence with two distinct roles. First, a production-scale *observation* from a companion empirical study [Dziuba et al., 2026]: end-to-end training of the spectral module loses badly to pretraining and freezing it. Second, a *controlled, instrumented study* on the same data and architecture family, designed to test the theorems' predictions directly — with matched arms, logged gradient dynamics, and the optimizer placed inside and outside the theorems' regime. The picture that emerges is sharper than simple confirmation: the block-scalar reading of Theorem 14 does *not* manifest at initialization, for a reason we can measure (rank-one input statistics); a direction-wise reading does manifest, strongly; and Theorem 27's functional prediction holds in its own regime while joint training outside that regime is *worse* than the theorem predicts — the spectral module is not merely starved but actively degraded.

8.1 Pipeline and capacity

The data are 336×336 pixel cores collected by QCL spectroscopy over the 1000–1800 cm^{-1} fingerprint region. Each pixel carries an $S = 942$ dimensional spectrum (after stacking raw absorbance and its first two derivatives). The classification target is per-pixel tissue type. The compositional model is the Block Vision Transformer of O'Leary et al. O'Leary et al. [2026]: a linear per-pixel $f_{\boldsymbol{\theta}} : \mathbb{R}^{942} \to \mathbb{R}^{K}$, then 16×16 patch tokenization, a 12-layer / 12-head Vision Transformer of embedding width $M = 192$, a transposed-convolution decoder, and a convolutional segmentation head. Parameter counts obtained directly from the instantiated model are given in Table 2.

Table 2: Module capacities of the BlockViT-v2 pipeline, counted from the instantiated model. K is the spectral bottleneck width; the two rows are the two configurations used in the companion's production runs.

Configuration	C_f	C_g	C_g/C_f
$K = 64$ (end-to-end baseline)	61,108	18,271,540	299
$K = 128$ (matched to frozen variants)	121,588	21,472,564	177

Channel convention. O'Leary et al.'s original BlockViT operates on the second-derivative spectrum alone (a single channel, $S = 965$ for their prostate FTIR data)—the field-standard preprocessing in infrared histopathology. The companion benchmark, and hence the models analyzed here, instead stacks raw absorbance with its first two derivatives (three channels, $S = 942$). The choice affects only C_f: instantiating the same model on the single-channel field-standard input gives $C_f = 20{,}916$ and $C_g/C_f \approx 874$. The three-channel convention is therefore the *conservative* choice for this paper's purposes—under the field standard, the capacity asymmetry we analyze is roughly three times larger.

The capacity ratio is thus roughly two orders of magnitude, and the spatial module's dominant blocks—the transformer (5.34×10^6) and the transposed convolution (9.44×10^6)—are both $\Theta(M^2)$ in the embedding width, so this architecture falls in the class treated by Theorem 14. Whether the curvature disparity the theorem permits actually manifests at initialization is an empirical question, which we now measure directly.

8.2 Production-scale observation

The companion study trains the full pipeline under 5-fold cross-validation on breast QCL data and reports held-out test macro-F1 for an end-to-end learned linear spectral module against three pretrain-and-freeze variants (its Table 6):

Two remarks on scope. First, this is an *observation that motivates the controlled study below*, not a theorem test: the variants differ in both stability (frozen vs. jointly trained) and informa-

Variant	Test F1 (5-fold mean $\pm$ sd)
End-to-end learned linear (joint)	0.675 ± 0.036
Frozen pretrained spectral (Peak Windows)	0.842 ± 0.121
Frozen pretrained spectral (MLP)	0.896 ± 0.092
Frozen pretrained spectral (Sliding Win.)	0.954 ± 0.020

Table 3: Held-out test F1 on breast QCL tissue cores, as published in the companion study [Dziuba et al., 2026]. End-to-end training of the spectral module underperforms every pretrain-and-freeze variant by 0.17–0.28 F1.

tiveness (pretrained vs. learned from scratch), and the production runs log no gradient dynamics. Second, the theorems do not guarantee that pretrain-and-freeze wins. Theorem 27's falsifiable prediction is that joint training behaves like *freezing the spectral module at initialization*; pretraining is a separate, empirically supported escape route. The controlled study disentangles exactly these factors.

8.3 Geometry at initialization

We measure the Gauss–Newton blocks of the instantiated pipeline on real training batches at random initialization (Lanczos with full reorthogonalization on matrix-free GGN block operators, validated to 10^{-12} against dense computation; Section 7), sweeping the ViT embedding width $M \in \{48, 96, 192, 384\}$ with 3 seeds $\times$ 4 batches per width.

The block-scalar disparity does not manifest. At production width $M = 192$ the measured curvature ratio is $\mathcal{D}_{\text{curv}} = \lambda_{\max}(G_{\boldsymbol{\phi\phi}})/\lambda_{\max}(G_{\boldsymbol{\theta\theta}}) = 0.46 \pm 0.21$ — the *spectral* block carries the larger top curvature, the opposite of the naive block-scalar expectation. The ratio rises with width (0.33 ± 0.12 at M=48 to 0.70 ± 0.19 at M=384; on prostate QCL it reaches parity, 1.09 ± 0.52, at M=384) but the disparity regime is nowhere in sight at buildable widths. Both theorem ingredients behave exactly as proved: the spectral cap is width-independent as Theorem 14 requires (log-log slope of $\lambda_{\max}(G_{\boldsymbol{\theta\theta}})$ vs. M: -0.003 breast, -0.012 prostate), and $\lambda_{\max}(G_{\boldsymbol{\phi\phi}})$ grows with M, though sublinearly over this range ($115 \to 258$ across an $8\times$ width increase). What fails is the *informativeness* of the bound, and the data tell us why.

Diagnosis: rank-one input statistics. The input second-moment matrix Σ_X of real tissue spectra has effective rank 1.07 (breast) and 1.02 (prostate) out of 942: every tissue spectrum shares the same dominant mean shape. The cap $C_\theta \propto \lambda_{\max}(\Sigma_X)$ is therefore enormous — the inequality of Theorem 14 holds but is vacuous at buildable widths, because the data concentrate all their second-moment mass in one direction. This is a measurable regime violation, not a counterexample: the theorem's disparity is a statement about what capacity *permits*, and rank-one inputs hand the spectral block a single direction of huge curvature.

Starvation is direction-wise. Precisely because the spectral curvature is rank-deficient, the block-scalar ratio is the wrong lens. Restricting the GGN to informative directions shows the disparity the scalar hides: the spectral block's top eigenvector aligns 0.954 with the mean-spectrum direction, and the top 40 of 61,108 spectral directions hold 83% of the trace. Along the clinically decisive CancerEpi–CAS class-contrast direction, the spectral curvature is 12–28$\times$ *smaller* than along the mean-spectrum direction; the spatial block's top eigenvalue overtakes the spectral spectrum already at rank 2–5. The gradient signal available for learning discriminative spectral features — as opposed to re-encoding the mean spectrum — is starved in exactly the directions that matter.

Normalization sets the cap. The cap is not an immutable property of the tissue: removing the spectral normalization deflates $\lambda_{\max}(G_{\theta\theta})$ by $2.4\times$ ($471 \to 194$ at M=192) and pushes the ratio across parity (1.18 at M=192, 1.31 at M=384). Input *centering*, by contrast, changes nothing when the architecture contains internal batch normalization ($\mathcal{D}_{\text{curv}}$ identical to three decimals with and without centering): a linear projection followed by normalization absorbs constant input shifts exactly. The lever on the curvature geometry acts through the normalization choice, not the preprocessing pipeline.

8.4 Controlled retraining dynamics

We then train the pipeline ourselves on breast QCL (fold 0), with full instrumentation: arms {joint-linear, frozen-random, frozen-PCA, joint-MLP} $\times$ widths $M \in \{48, 192\} \times 3$ seeds under AdamW (60 epochs), plus a {joint-linear, frozen-random} $\times$ 3-seed pair under plain SGD at M=48 — the closest buildable match to the theorems' regime. Following Section 5, the spectral parameters sit in a zero-weight-decay group in every arm; frozen arms log counterfactual spectral gradients. Per-step residuals, block gradient norms, and per-epoch parameter displacements and input-Jacobian norms are recorded.

Table 4: Controlled study: held-out validation macro-F1 (mean $\pm$ sd over 3 seeds, mean of final 5 epochs) and relative spectral-parameter displacement at end of training, AdamW arms.

Arm	M=48	M=192	$\lVert\Delta\boldsymbol{\theta}\rVert/\lVert\boldsymbol{\theta}_0\rVert$
Joint (linear)	0.538 ± 0.082	0.646 ± 0.050	0.55–0.60
Frozen random	0.648 ± 0.063	0.731 ± 0.023	0
Frozen PCA	0.731 ± 0.003	0.727 ± 0.012	0
Joint (MLP)	0.566 ± 0.070	0.575 ± 0.184	0.86–0.95

Frozen beats joint everywhere. Under AdamW, freezing the spectral module at a *random* initialization outperforms training it jointly at both widths (gaps -0.110 at M=48, -0.085 at M=192). The pre-registered falsifier — joint beating frozen-PCA decisively — comes out negative at both widths (-0.193, -0.081). The production-scale observation of Section 8.2 survives the controlled setting with informativeness held fixed: *stability alone* accounts for a large fraction of the gap. Indeed at M=192 frozen-PCA and frozen-random are statistically indistinguishable (0.727 vs. 0.731); the informativeness of the frozen features matters at small width (0.731 vs. 0.648 at M=48) but stability is what the theory speaks to, and stability is what does the work here.

Harmful drift, not lazy pinning. The joint arms' spectral parameters are not starved into stillness: they move 55–60% in relative norm under AdamW (86–95% for the MLP spectral module) — and this movement *hurts*, since the same module left at its random initialization scores higher. Outside the theorems' regime the shortcut mechanism is thus stronger than the starvation story: adaptive preconditioning renormalizes the small spectral gradients (Section 4.5) and the spectral module chases the residual left by the faster spatial module, degrading rather than refining its features. The nonlinear joint-MLP arm shows the same phenomenology, so the effect is not an artifact of the linear parameterization.

SGD dissociation: the theorem's regime behaves as proved. Under plain SGD with zero weight decay — the setting of Theorem 27 — the story reverts to the theorem's: joint training and frozen-random become statistically indistinguishable (0.538 ± 0.075 vs. 0.510 ± 0.051; gap $+0.028$, within seed noise), which is precisely the theorem's functional-equivalence prediction, and the spectral share of total squared gradient flow drops from ≈ 0.94 (AdamW joint) to ≈ 0.27. The optimizer is the dissociating variable: the flow theorems describe SGD-like dynamics correctly, and adaptive preconditioning is what carries the dynamics out of their scope — consistent with the preconditioned-spectrum discussion in Section 4.5.

Envelope curvature grows with width. The fitted residual-envelope rate for the joint-linear arm grows from $\hat{\mu} = 6.0 \times 10^{-4}$ (M=48) to 1.9×10^{-3} (M=192), a factor 3.2 for a 4× width increase — qualitatively consistent with the width-growing NTK floor of Proposition 31, which is the theory's proved mechanism for why the spatial module's fast fit accelerates with capacity.

The scalar EGR diagnostic is blind here. The scalar effective-gradient-ratio *rises* during joint AdamW training (epoch-1 median ≈ 0.8 to ≈ 4 by epoch 60) even as the jointly trained module underperforms its own frozen initialization. A single global ratio cannot see direction-wise starvation (cf. Remark 35): the spectral gradients are large along the mean-spectrum direction and starved along the discriminative ones, exactly as the initialization geometry of Section 8.3 predicts. Direction-resolved diagnostics are future work.

8.5 What the theorems do and do not predict here

Honest scoping of the quantitative statements. The containment assumption of Theorem 27 is violated on real data: the spatial module's input-Jacobian operator norm grows by orders of magnitude over training ($3.2 \to 157$ on average for joint-linear, up to 414; $1.7 \to 1.2 \times 10^4$ for joint-MLP). Instantiating the theorem's displacement bound with measured constants accordingly yields a bound 2×10^4–2×10^5 times the measured displacement: on this pipeline the bound is true but vacuous, and we report it as such. The synthetic setting of Section 7, where containment holds by construction, is where the bound is quantitatively informative; the real-data value of the theory lies in its verified ingredients (width-independent spectral cap, width-growing spatial curvature, the class-prior NTK floor) and in the regime-resolved dynamics above — the functional equivalence it predicts appears exactly where its assumptions apply, and the failure outside that regime is worse for joint training, not better.

The companion study [Dziuba et al., 2026] additionally provides a pixel-level tokenization benchmark across three datasets and the spatial-transfer experiment on breast QCL from which Table 3 is drawn; those results characterize *what* performs well, while the present analysis addresses *why* end-to-end training of the spectral stage underperforms.

9 Discussion

9.1 Implications for practice

The theorems and experiments establish a concrete prescription. In any compositional spectral-spatial pipeline with substantial capacity asymmetry $C_g \gg C_f$, end-to-end joint training is expected to fail in a specific and predictable way: the spectral module stays near initialization while the spatial module fits the data using shortcuts in the unstructured feature map. Three operational responses follow.

Freeze the spectral module. The most direct response to Theorem 27 is to remove $\boldsymbol{\theta}$ from the optimization. Without a spectral update, the timescale separation is eliminated by construction, and $g_{\boldsymbol{\phi}}$ optimizes a standard supervised learning problem on fixed input features. If a pretrained spectral encoder is available (from unsupervised pretraining, self-supervised pretraining, or pre-task pretraining on the same data), it should be used frozen.

Use a fixed baseline. When pretraining is unavailable, a fixed but informative baseline $h(X)$ (e.g. PCA, fixed wavelet basis, random projection) can be injected as an additive component $Z = f_{\boldsymbol{\theta}}(X) + h(X)$. The fixed component ensures the spatial model receives a stable input even if $f_{\boldsymbol{\theta}}$ drifts; the additive structure allows fine-grained refinement of $f_{\boldsymbol{\theta}}$ without exposing the spatial model to the moving-target dynamics that produce the shortcut. We have not empirically tested this prescription in this paper; it follows directly from the theory and is a natural focus for follow-up work.

Track EGR. The EGR diagnostic detects the onset of starvation in real time. Practitioners should log it alongside training and validation loss. When $\mathrm{EGR}(t)$ drops below a chosen threshold (we recommend $0.1 \cdot \mathrm{EGR}(0)$), the theoretical analysis predicts that subsequent $\boldsymbol{\theta}$ updates are essentially noise: the spectral module has been absorbed into a spurious stationary point. At this point, freezing $\boldsymbol{\theta}$ and continuing to train $\boldsymbol{\phi}$ retains the run's investment while avoiding the negative consequences of further joint updates.

9.2 Limitations

We make four limitations explicit.

Linear spectral reduction. Theorems 14 and 27 are proven under the assumption that $f_{\boldsymbol{\theta}}$ is linear in $\boldsymbol{\theta}$. In practice some pipelines use shallow nonlinear spectral modules (1–2 layer MLPs). The intuition behind both theorems extends: nonlinear shallow $f_{\boldsymbol{\theta}}$ retains the property that the spectral block carries substantially less capacity than the spatial block, so the qualitative timescale separation persists. We verify empirically in Section 7 that the joint-vs-frozen gap remains under nonlinear $f_{\boldsymbol{\theta}}$. A formal proof for nonlinear $f_{\boldsymbol{\theta}}$ is left to future work.

Finite-width scaling exponent. The empirical scaling exponent of λ_ϕ vs. C_g is 0.70 ± 0.05 at the widths we tested, while the width-linear prediction of Lemma 16 for the tested architecture family is 1.0 (the SpatialMLP has $C_g = \Theta(M)$, so slope 1.0 applies against C_g as well as against M). The measurement is sublinear—it undershoots the asymptote. Whether the asymptote is approached at larger widths, or the deficit reflects the measurement and idealization gaps discussed in Section 7, is left open. Two consequences for the paper: the Theorem 14 bound should be read as a lower bound that may be loose at finite scale, and the empirical predictions should be interpreted as qualitative (monotonic growth) rather than tight (constant proportionality).

Memorization regime. For sufficiently overparametrized spatial models (ViT in our experiments), joint training enters a memorization regime in which train loss falls to near zero and both gradient norms approach the noise floor. In this regime the EGR observable becomes uninformative, even though the underlying prescription (freeze) continues to apply. Practitioners running highly overparametrized models should observe the symptoms (test loss divergence, train loss saturation) rather than relying on EGR alone.

Single domain validation. Our real-data validation is restricted to FTIR/QCL tissue classification in breast cancer. We expect the same mechanism to operate in hyperspectral remote sensing, Raman tissue classification, and mass spectrometry imaging, but those validations are future work.

9.3 Defense against the "linear triviality" critique

A natural reviewer objection is that, because $f_{\boldsymbol{\theta}}$ is linear, the analysis must reduce to principal component analysis. We address this in Remark 22 and emphasize again: PCA finds directions of maximum input variance independently of labels. Theorem 14 concerns curvature of the supervised cross-entropy loss, which depends on the current state of $g_{\boldsymbol{\phi}}$ through the Jacobian $\partial\hat{y}/\partial Z$. The pathology we identify is that this supervised signal fails to reach $\boldsymbol{\theta}$. A PCA-decomposed $f_{\boldsymbol{\theta}}$ would not exhibit the joint-vs-frozen gap we observe: indeed, our experiments include a frozen-PCA baseline that performs well, exactly because PCA decomposition does not require the supervised signal to flow through it.

9.4 The Spatial Dominance Conjecture: an open problem

A stronger statement than Theorem 27 is suggested by our experiments but not proven: *when spectral and spatial features carry equal information about the label, joint training exhibits an implicit bias toward the spatial pathway.* Pezeshki et al. Pezeshki et al. [2021] prove a related but distinct result in the NTK regime; Huang et al. Huang et al. [2022] prove a related result for explicit dual-modality networks but not for compositional single-modality pipelines. A proof of the spatial-dominance conjecture in our setting would require formalizing "equal information" in an operationally meaningful way (Shannon mutual information is bijection- invariant and therefore not the right primitive for gradient dynamics) and combining the NTK eigenvalue gap with our compositional Hessian analysis. We leave this as an open problem.

9.5 Beyond spectroscopy: where else this matters

Although our motivating application is infrared tissue classification, the mechanism we identify applies to any compositional architecture with capacity asymmetry between sequential modules. We highlight three areas where the same pathology likely operates.

Hyperspectral remote sensing. The dominant paradigm for hyperspectral image classification is a shallow spectral reduction (PCA, autoencoder, or 1D conv) feeding into a deep spatial classifier (3D CNN, ViT). The capacity asymmetry is typically large. Joint training is the de facto standard. Our analysis predicts the same shortcut emergence; whether the field has noticed it under the guise of "classification works without much spectral processing" is a question worth examining.

Multimodal foundation models. Vision-language and audio-language models compose a sequence of modality-specific encoders into a shared representation space. The modality with the larger encoder typically dominates the shared representation. While this has been documented empirically (Wang et al. Wang et al. [2020], Peng et al. Peng et al. [2022]), the mechanism is the same as ours: capacity-induced timescale separation combined with cross-entropy saturation. Our framework extends their empirical observations into the single-input compositional setting.

Video models with frozen image encoders. A standard practice in video understanding is to freeze a pretrained image encoder and train a temporal model on top. This is widely treated as "transfer learning," but the same mechanism we describe predicts that joint training would fail in a specific way – the image encoder would drift, the temporal model would absorb the shortcut, generalization would suffer. The empirical literature strongly supports the freezing practice; our analysis provides a mechanistic reason.

9.6 Concluding remarks

We have provided the first rigorous explanation for the failure of end-to-end joint training in compositional spectral-spatial deep learning. The phenomenon is geometric (Theorem 14) and dynamical (Theorem 27), is detectable in real time via the EGR diagnostic, and has a direct prescriptive remedy (freezing or fixed-baseline injection). We expect the framework to extend beyond spectroscopy to any compositional architecture with capacity asymmetry, and to provide a principled foundation for the empirical practice of frozen feature extractors that the field has converged on by necessity rather than design.

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A Supplementary Material

This supplement provides the full proofs of Theorems 14 and 27, additional experiments referenced in the main text, and robustness checks.

A.1 Full proof of Theorem 14

We restate Theorem 14 for reference:

> *Under Assumptions 1, 4, and 7, together with the head hypothesis of Lemma 16, the module curvature disparity satisfies, in expectation over random Kaiming (variance-scaled Gaussian) initialization, $\mathcal{D}_{\mathrm{curv}}^{(M)} \geq c_1 \cdot M - c_2$, and for architecture families with $C_g = \Theta(M^2)$, $\mathcal{D}_{\mathrm{curv}} \geq c_1' \sqrt{C_g/C_f} - c_2$, for constants depending on the data distribution, the initialization scheme, the bottleneck dimension K, the input spectral dimension S, and the architecture class of g_ϕ—but not on M or C_g.*

The proof rests on two lemmas: (i) the spatial block's top eigenvalue scales linearly with width M (Lemma 16, proved by an explicit witness direction in the head weights, with Karakida et al.'s mean-field law as the sharp asymptotic benchmark); (ii) the spectral block's top eigenvalue is capped by a C_g-independent operator-norm bound (Lemma 18). We prove each lemma in turn, then combine.

A.1.1 Proof of Lemma 16

We show $\lambda_{\max}(G_{\phi\phi}) = \Omega(M)$ in expectation over random Kaiming (variance-scaled Gaussian) initialization, under the head hypothesis of Lemma 16: the classifier head of g_ϕ is a dense layer over penultimate features $h_n \in \mathbb{R}^{M_h}$ with $M_h = \Theta(M)$ whose softmax-weighted mean Gram has top eigenvalue $\Omega(M_h)$ at initialization. The proof has three steps: (1) context—Karakida et al.'s exact width-linear law for the top Fisher eigenvalue in their model class, which we use as the asymptotic benchmark but do *not* invoke directly; (2) an exact spectral lower bound on the softmax inner matrix at initialization; (3) an elementary explicit witness direction in the head weights that yields $\Omega(M)$ self-containedly. Throughout, Grams are normalized as empirical means over the $N := N_{\mathrm{data}}HW$ pixel positions; the unnormalized (summed) Grams differ by the fixed factor N, which does not affect any statement in M.

Step 1: The width-linear law of Karakida et al. (context). Karakida, Akaho, and Amari Karakida et al. [2019] analyze fully connected networks of depth $L \geq 2$ with layer widths $M_\ell = \alpha_\ell M$, linear outputs, and variance-scaled Gaussian (Kaiming-type) initialization $W_{ij}^\ell \sim \mathcal{N}(0, \sigma_w^2/M_{\ell-1})$, $b_i^\ell \sim \mathcal{N}(0, \sigma_b^2)$. For the empirical Fisher information of the squared-loss model, $F = \frac{1}{T}\sum_{t=1}^{T}\sum_k \nabla f_k(x_t)\nabla f_k(x_t)^\top$ (the empirical-mean Gram of the output Jacobian), their Theorem 4 gives, in the limit $M \gg 1$,

$$\lambda_{\max}(F) \;=\; \alpha\left(\tfrac{T-1}{T}\,\kappa_2 + \tfrac{1}{T}\,\kappa_1\right) M \;=\; \Theta(M),$$

where $\kappa_1, \kappa_2 = \Theta(1)$ are macroscopic constants computed from the layer-wise signal-propagation recursions and $\alpha = \sum_\ell \alpha_\ell \alpha_{\ell-1}$. Their Theorem 1 simultaneously shows the *mean* eigenvalue is $m_\lambda = C\kappa_1/M = \mathcal{O}(1/M)$: as the network widens, the FIM spectrum becomes extremely skewed—almost all directions flatten while the top eigenvalue grows linearly in width. We record two cautions. First, Theorem 4 fixes the *output* layer at $C = \mathcal{O}(1)$ but requires the input and hidden widths to scale proportionally with M; a classifier reading a fixed bottleneck K violates the input-side requirement (its coefficient α degenerates there), so the theorem cannot be applied verbatim to our spatial modules. Second, both statements are asymptotic mean-field equalities with no finite-M rate. Steps 2–3 therefore prove the $\Omega(M)$ bound self-containedly; Karakida's law serves as the sharp benchmark within its model class, validated numerically in their Figure 1.

Step 2: The softmax inner matrix at initialization. Our loss is softmax cross-entropy, not squared loss, so the spatial Gauss–Newton block carries the softmax inner matrix

$$G_{\phi\phi} \;=\; \frac{1}{N}\sum_{n=1}^{N} J^{(n)\top} H_\tau^{(n)} J^{(n)}, \qquad H_\tau \;:=\; \mathrm{diag}(p) - pp^\top,$$

where $J^{(n)} := \partial \hat{y}^{(n)}/\partial \phi$ is the per-pixel logit Jacobian and $p = \mathrm{softmax}(\hat{y})$. For any $v \in \mathbb{R}^{N_{\mathrm{cls}}}$, the quadratic form is a variance: $v^\top H_\tau v = \sum_i p_i v_i^2 - (\sum_i p_i v_i)^2 = \mathrm{Var}_{i\sim p}(v_i) \ge p_{\min} \sum_i (v_i - \bar{v}_p)^2 \ge p_{\min} \|P_\perp v\|^2$, where $p_{\min} := \min_i p_i$, $\bar{v}_p := \sum_i p_i v_i$, the last inequality uses that the unweighted mean minimizes the sum of squared deviations, and $P_\perp := I - \frac{1}{N_{\mathrm{cls}}}\mathbf{1}\mathbf{1}^\top$ projects onto the mean-zero subspace $\mathbf{1}^\perp$. Hence

$$H_\tau \;\succeq\; p_{\min} P_\perp.$$

The bound is used per pixel in Step 3; we do not take a minimum over pixels, so no uniform-over-pixels lower bound on $p_{\min}$ is needed anywhere in the proof. (For orientation: at Kaiming initialization the logits are $\mathcal{O}(1)$ uniformly in M, so $p_{\min}^{(n)}$ is of order $1/N_{\mathrm{cls}}$ for typical n.)

Step 3: An explicit witness direction. Write the classifier head of g_ϕ as $\hat{y}_n = W h_n + b$ with penultimate features $h_n = h(x_n) \in \mathbb{R}^{M_h}$, $M_h = \Theta(M)$. A perturbation of the head weights is a matrix $V \in \mathbb{R}^{N_{\mathrm{cls}} \times M_h}$; it is a unit vector in parameter space precisely when $\|V\|_F = 1$, under any vectorization convention, and the logit perturbation it induces at pixel n is $V h_n$. Define the *softmax-weighted feature Gram*

$$S_h^p \;:=\; \frac{1}{N}\sum_{n=1}^{N} p_{\min}^{(n)}\, h_n h_n^\top, \qquad p_{\min}^{(n)} := \min_i p_i^{(n)}, \tag{11}$$

and let u be a top unit eigenvector of S_h^p. Fix a unit contrast $c \in \mathbb{R}^{N_{\mathrm{cls}}}$ with $c \perp \mathbf{1}$ (possible since $N_{\mathrm{cls}} \ge 2$); by Step 2, at every pixel

$$c^\top H_\tau^{(n)} c \;=\; \mathrm{Var}_{i\sim p^{(n)}}(c_i) \;\ge\; p_{\min}^{(n)} \|P_\perp c\|^2 \;=\; p_{\min}^{(n)}.$$

Take the rank-one witness $V := c\, u^\top$, for which $\|V\|_F = \|c\|\,\|u\| = 1$ and $V h_n = (h_n^\top u)\, c$. Then

$$\lambda_{\max}(G_{\phi\phi}) \;\ge\; \frac{1}{N}\sum_{n=1}^{N} (V h_n)^\top H_\tau^{(n)} (V h_n) \;=\; \frac{1}{N}\sum_{n=1}^{N} (h_n^\top u)^2\, c^\top H_\tau^{(n)} c \;\ge\; u^\top S_h^p u \;=\; \lambda_{\max}(S_h^p). \tag{12}$$

Note what (12) is: a deterministic, hypothesis-free inequality—one identity and one application of Step 2—valid at every parameter value. All N pixels contribute, and no minimum over pixels is taken, so no union bound and no factor of N enters.

The head hypothesis of Lemma 16 is exactly $\lambda_{\max}(S_h^p) \ge q_h M_h$ with probability at least $1 - \delta_h$ over initialization, for $q_h > 0$ and $\delta_h < 1$ independent of M. On that event $\lambda_{\max}(G_{\phi\phi}) \ge q_h M_h = \Omega(M)$; since $\lambda_{\max} \ge 0$ always,

$$\mathbb{E}[\lambda_{\max}(G_{\phi\phi})] \;\ge\; (1 - \delta_h)\, q_h M_h \;=\; \Omega(M),$$

which is the claim.

On the head hypothesis, and its counterpart in Karakida et al. Write $S_h := N^{-1} \sum_n h_n h_n^\top$ for the unweighted mean feature Gram. The hypothesis $\lambda_{\max}(S_h^p) = \Omega(M_h)$ asks two things at once: that the penultimate features carry a common direction of $\Theta(M_h)$ energy across inputs ($\lambda_{\max}(S_h) = \Omega(M_h)$), and that the initial predictions are non-degenerate on the pixels supplying that energy. The second is mild at initialization—the logits are $\mathcal{O}(1)$ uniformly in M, since the head weights have variance $\Theta(1/M_h)$ against $\|h_n\|^2 = \mathcal{O}(M_h)$, so $p_{\min}^{(n)}$ is bounded below by a constant of order $1/N_{\text{cls}}$ on all but a small fraction of pixels, and S_h^p dominates that constant times the unweighted Gram restricted to those pixels. (Proposition 38 below turns this sketch into a proof for a concrete head.) The first is the substantive condition. It is implied by $\|\bar{h}\|^2 \geq qM_h$ on the *mean* feature vector $\bar{h} := N^{-1} \sum_n h_n$: for $\bar{u} = \bar{h}/\|\bar{h}\|$,

$$\lambda_{\max}(S_h) \;\geq\; \bar{u}^\top S_h \bar{u} \;=\; \frac{1}{\|\bar{h}\|^2} \cdot \frac{1}{N} \sum_n (h_n^\top \bar{h})^2 \;\geq\; \frac{1}{\|\bar{h}\|^2} \left(\frac{1}{N} \sum_n h_n^\top \bar{h} \right)^2 \;=\; \frac{\|\bar{h}\|^4}{\|\bar{h}\|^2} \;=\; \|\bar{h}\|^2,$$

the middle step being Jensen's inequality (the mean of squares dominates the square of the mean). At variance-scaled initialization each entry of h_n has $\Theta(1)$ second moment (the signal-propagation recursions of Karakida et al. Karakida et al. [2019], Eq. (9)), so the condition holds whenever the activations are correlated across inputs—in particular for any nonnegative or nonzero-mean activation (ReLU and its variants), and for odd activations whenever the bias variance $\sigma_b^2 > 0$ or the inputs are themselves correlated.

Note that per-entry second moments alone are *not* enough: $\|h_n\|^2 = \Theta(M_h)$ for every n gives only $\operatorname{tr}(S_h) = \Theta(M_h)$, hence $\lambda_{\max}(S_h) \geq \Theta(M_h)/\min(N, M_h)$, which is weaker by a factor of N. Cross-input correlation, not per-vector norm, is what makes the top eigenvalue grow.

The same quantity carries the width-linear term in Karakida et al.'s Theorem 4, $\lambda_{\max} = \alpha(\frac{T-1}{T}\kappa_2 + \frac{1}{T}\kappa_1)M$: their κ_2 is built from the cross-input activation correlations $\hat{q}_{st}$. Their footnote notes that odd activations with $\sigma_b = 0$ give $\hat{q}_{st} = 0$ and $\kappa_2 = 0$, and is explicit that this exceptional case falls outside their analysis: "In such exceptional cases, we need to evaluate the lower order terms of s_λ and $\lambda_{\max}$ (outside the scope of this study)." We therefore attribute no finite-T claim to their theorem in the degenerate case. Our hypothesis has the parallel structure, and its degenerate behavior *can* be evaluated directly: for odd activations with $\sigma_b = 0$ and uncorrelated inputs the features lose their common direction, and the per-vector norms alone give only $\lambda_{\max}(S_h) \geq \operatorname{tr}(S_h)/\min(N, M_h)$. In our regime $N \gg M_h$, so this floor is $\Theta(1)$—*no* width scaling. Width-linear growth in the degenerate case would require a separate argument, which we do not attempt—consistent with Karakida et al.'s footnote placing that case outside their scope as well. For CNN and ViT heads we state the hypothesis as an explicit assumption rather than a theorem, and check it empirically in Section 7.

Proposition 38 (A fully proved instance)**.** *Fix bottleneck features $z_1, \ldots, z_N \in \mathbb{R}^K$ with $\|z_n\| \leq R$, and let the spatial module's head be the two-layer ReLU network*

$$h_n = \operatorname{ReLU}(W_1 z_n + b_1), \qquad \hat{y}_n = W_2 h_n + b_2,$$

with Kaiming Gaussian initialization: entries of $W_1 \in \mathbb{R}^{M_h \times K}$ i.i.d. $\mathcal{N}(0, \sigma_w^2/K)$, entries of $W_2 \in \mathbb{R}^{N_{\text{cls}} \times M_h}$ i.i.d. $\mathcal{N}(0, \sigma_w^2/M_h)$, biases i.i.d. $\mathcal{N}(0, \sigma_b^2)$ with $\sigma_b > 0$, and $N_{\text{cls}} \geq 2$. Let the spectral module be the linear per-pixel $f_{\boldsymbol{\theta}}$ of Assumption 1 with inputs satisfying Assumption 4. Then there exist constants $q_h > 0$, $M_0 < \infty$, and $C_G < \infty$, depending only on $\sigma_w, \sigma_b, R, K, N_{\text{cls}}$, and $\operatorname{tr}(\Sigma_X)$—not on M_h or N—such that for all $M_h \geq M_0$, at initialization:

(i) (head hypothesis) $\lambda_{\max}(S_h^p) \geq q_h M_h$ with probability at least $1/2$; consequently $\mathbb{E}[\lambda_{\max}(G_{\boldsymbol{\phi\phi}})] \geq \frac{1}{2} q_h M_h$;

(ii) (spectral-block cap, trace form) $\mathbb{E}[\lambda_{\max}(G_{\boldsymbol{\theta\theta}})] \leq \mathbb{E}[\operatorname{tr}(G_{\boldsymbol{\theta\theta}})] \leq C_G := N_{\text{cls}}\, \sigma_w^4 \operatorname{tr}(\Sigma_X)$, uniformly in M_h and N;

(iii) (instance-level disparity law) with probability at least 3/8, $\mathcal{D}_{\rm curv} \geq (q_h/(8C_G))\, M_h$; consequently $\mathbb{E}[\mathcal{D}_{\rm curv}] \geq \frac{3}{64}(q_h/C_G)\, M_h = \Omega(M)$.

For this architecture, the disparity law of Theorem 14—not merely its hypotheses—is proved outright at initialization. (What remains assumption-level in general is the uniform-over-inputs, during-training form of Assumption 7: a per-input Jacobian cap at initialization follows from the computation in Step D below, but a supremum over all inputs by the natural union route would cost a $\sqrt{\log N}$ factor, and training-time persistence is exactly what Assumption 7 postulates. The trace-form cap (ii) bypasses both issues for the instance.)

Proof. Step A (mean feature vector). Write $g_{n,i} = w_i^\top z_n + b_{1,i}$ for the pre-activations, so $g_{n,i} \sim \mathcal{N}(0, s_n^2)$ with $s_n^2 = \sigma_w^2 \|z_n\|^2 / K + \sigma_b^2 \in [\sigma_b^2, s_{\max}^2]$, $s_{\max}^2 := \sigma_w^2 R^2/K + \sigma_b^2$. Since $\mathbb{E}\,\mathrm{ReLU}(\mathcal{N}(0, s^2)) = s/\sqrt{2\pi}$, each coordinate of the mean feature vector $\bar{h} = N^{-1}\sum_n h_n$ has expectation

$$m \;:=\; \mathbb{E}[\bar{h}_i] \;=\; \frac{1}{N}\sum_n \frac{s_n}{\sqrt{2\pi}} \;\geq\; \frac{\sigma_b}{\sqrt{2\pi}} \;>\; 0.$$

To obtain constants free of the data and of N, we use throughout only the uniform floor $m \geq m_0 := \sigma_b/\sqrt{2\pi}$. The rows $(w_i, b_{1,i})$ are i.i.d. across i, so the coordinates $\bar{h}_i$ are i.i.d. with $\mathbb{E}[\bar{h}_i^2] \leq N^{-1}\sum_n \mathbb{E}[\mathrm{ReLU}(g_{n,i})^2] \leq s_{\max}^2$ and (being averages of sub-Gaussian variables) finite fourth moment bounded by a constant $C_4(s_{\max})$. By Chebyshev's inequality applied to the i.i.d. sum $\|\bar{h}\|^2 = \sum_i \bar{h}_i^2$, whose mean is at least $m^2 M_h \geq m_0^2 M_h$,

$$\Pr\Big[\|\bar{h}\|^2 < \tfrac{1}{2} m_0^2 M_h\Big] \;\leq\; \frac{4\,C_4(s_{\max})}{m_0^4\, M_h} \;\xrightarrow[M_h\to\infty]{}\; 0.$$

On the complementary event E_A, the Jensen step above gives $\lambda_{\max}(S_h) \geq \|\bar{h}\|^2 \geq \frac{1}{2} m_0^2 M_h$, and moreover the witness direction $\bar{u} = \bar{h}/\|\bar{h}\|$ certifies $A := N^{-1}\sum_n (h_n^\top \bar{u})^2 \geq \frac{1}{2} m_0^2 M_h$.

Step B (softmax weights). Condition on $\mathcal{F}_1 = \sigma(W_1, b_1)$, which fixes the features h_n, the direction $\bar{u}$, and the energies $a_n := (h_n^\top \bar{u})^2$; the head randomness (W_2, b_2) is independent of $\mathcal{F}_1$. Call pixel n *tame* if $\|h_n\|^2 \leq 2 s_{\max}^2 M_h$. For a tame pixel, the logits $\hat{y}_n \,|\, \mathcal{F}_1$ have i.i.d. Gaussian entries with variance $\sigma_w^2 \|h_n\|^2 / M_h + \sigma_b^2 \leq 2\sigma_w^2 s_{\max}^2 + \sigma_b^2 =: s_*^2$, so by a Gaussian tail bound on the logit range, there are constants $c_p(s_*, N_{\rm cls}) > 0$ and $\delta_0 \leq 1/16$ with $\Pr[p_{\min}^{(n)} < c_p \mid \mathcal{F}_1] \leq \delta_0$ for every tame pixel. Untame pixels are exponentially rare: $\|h_n\|^2 / M_h$ is an average of M_h i.i.d. sub-exponential variables with mean $\leq s_{\max}^2/2$, so $\Pr[n \text{ untame}] \leq e^{-cM_h}$ (Bernstein's inequality; Vershynin Vershynin [2018], Thm. 2.8.1), and by Cauchy–Schwarz with $\mathbb{E}\,\|h_n\|^4 \leq 3 s_{\max}^4 M_h^2$ (i.i.d. coordinates), the $\mathcal{F}_1$-measurable untame energy $U := N^{-1}\sum_n a_n \mathbf{1}[n \text{ untame}]$ satisfies

$$\mathbb{E}[U] \;\leq\; \max_n \sqrt{\mathbb{E}\,\|h_n\|^4}\sqrt{\Pr[n \text{ untame}]} \;\leq\; \sqrt{3}\, s_{\max}^2 M_h\, e^{-cM_h/2} \;=:\; \varepsilon(M_h)\, M_h,$$

with $\varepsilon(M_h) \to 0$ exponentially (we used $a_n \leq \|h_n\|^2$, valid since $\bar{u}$ is a unit vector).

Step C (assembly). We control the two bad masses by separate arguments, respecting which randomness each depends on. The untame energy U is $\mathcal{F}_1$-measurable; by (unconditional) Markov, the event $E_U := \{U \leq 16\,\varepsilon(M_h) M_h\}$ has $\Pr[E_U] \geq 15/16$, and for $M_h \geq M_0$ large enough that $16\,\varepsilon(M_h) \leq m_0^2/8$, on $E_A \cap E_U$ we have $U \leq \frac{1}{8} m_0^2 M_h \leq \frac{1}{4} A$. The p-degenerate mass over *tame* pixels, $P := N^{-1}\sum_{n \text{ tame}} a_n \mathbf{1}[p_{\min}^{(n)} < c_p]$, is (W_2, b_2)-random given $\mathcal{F}_1$, with $\mathbb{E}[P \mid \mathcal{F}_1] \leq \delta_0 A$ by Step B; conditional Markov with $\delta_0 \leq 1/16$ gives $\Pr[\,P > \frac{1}{4} A \mid \mathcal{F}_1\,] \leq 4\delta_0 \leq 1/4$. On the intersection of the three events—probability at least $1 - 4C_4/(m_0^4 M_h) - 1/16 - 1/4 > 1/2$ for $M_h \geq M_0$—the total bad mass is $P + U \leq \frac{1}{2} A$, and

$$\lambda_{\max}(S_h^p) \;\geq\; \bar{u}^\top S_h^p \bar{u} \;\geq\; c_p\,(A - P - U) \;\geq\; \frac{c_p}{2}\, A \;\geq\; \frac{c_p m_0^2}{4}\, M_h.$$

Take $q_h := c_p m_0^2/4 = c_p\sigma_b^2/(8\pi)$, manifestly free of the data and of N. The in-expectation clause follows from $\lambda_{\max} \geq 0$ and (12). This proves (i).

Step D (spectral-block cap, trace form). At any input z, the head's input Jacobian is $\partial\hat{y}/\partial z = W_2 D_z W_1$ with $D_z = \mathrm{diag}(\mathbf{1}[W_1 z + b_1 > 0])$. Conditioning on (W_1, b_1) (which fixes D_z) and using that the rows of W_2 are i.i.d. $\mathcal{N}(0, (\sigma_w^2/M_h)I)$,

$$\mathbb{E}\,\|W_2 D_z W_1\|_F^2 \;=\; N_{\text{cls}} \cdot \frac{\sigma_w^2}{M_h} \cdot \mathbb{E}\,\|D_z W_1\|_F^2 \;\leq\; N_{\text{cls}} \cdot \frac{\sigma_w^2}{M_h} \cdot \mathbb{E}\,\|W_1\|_F^2 \;=\; N_{\text{cls}}\,\sigma_w^4,$$

using $\mathbb{E}\,\|W_1\|_F^2 = M_h K \cdot \sigma_w^2/K = M_h\sigma_w^2$—uniformly over inputs z. (The naive product bound $\|W_2\|_{\text{op}}\,\|W_1\|_{\text{op}}$ would grow like $\sqrt{M_h/K}$; the composition is $O(1)$ because the top directions of the two random factors are generically misaligned—which the Frobenius computation captures without any alignment analysis.) Now bound the spectral block. With $J_{\boldsymbol{\theta}}^{(n)} = (W_2 D_{z_n} W_1)(I_K \otimes x_n^\top)$ the per-pixel spectral logit Jacobian (Lemma 18, Step 3) and $H_\tau \preceq I$,

$$\mathbb{E}\,\mathrm{tr}(G_{\boldsymbol{\theta}\boldsymbol{\theta}}) \;\leq\; \frac{1}{N}\sum_{n=1}^{N} \mathbb{E}\left\|J_{\boldsymbol{\theta}}^{(n)}\right\|_F^2 \;\leq\; \frac{1}{N}\sum_{n=1}^{N} \mathbb{E}\,\|W_2 D_{z_n} W_1\|_F^2 \cdot \|x_n\|^2 \;\leq\; N_{\text{cls}}\,\sigma_w^4 \cdot \frac{1}{N}\sum_{n}\|x_n\|^2 \;=\; N_{\text{cls}}\,\sigma_w^4\,\mathrm{tr}(\Sigma_X),$$

using $\|AB\|_F \leq \|A\|_F\,\|B\|_{\text{op}}$ and $\left\|I_K \otimes x_n^\top\right\|_{\text{op}} = \|x_n\|$. Since $\lambda_{\max} \leq \mathrm{tr}$ for positive semidefinite matrices, (ii) follows with $C_G = N_{\text{cls}}\sigma_w^4\mathrm{tr}(\Sigma_X)$, free of M_h and N.

Step E (combining). If $C_G = 0$ then $\Sigma_X = 0$, the spectral block vanishes almost surely, and $\mathcal{D}_{\text{curv}} = +\infty$ by the convention of Definition 12; assume $C_G > 0$ below. Let E_1 be the event of (i) ($\Pr \geq 1/2$) and $E_2 := \{\lambda_{\max}(G_{\boldsymbol{\theta}\boldsymbol{\theta}}) \leq 8C_G\}$, which by (ii) and Markov has $\Pr[E_2] \geq 7/8$. On $E_1 \cap E_2$ (probability at least $1 - 1/2 - 1/8 = 3/8$), the witness bound (12) gives

$$\mathcal{D}_{\text{curv}} \;=\; \frac{\lambda_{\max}(G_{\boldsymbol{\phi}\boldsymbol{\phi}})}{\lambda_{\max}(G_{\boldsymbol{\theta}\boldsymbol{\theta}})} \;\geq\; \frac{q_h M_h}{8C_G},$$

and since $\mathcal{D}_{\text{curv}} \geq 0$, $\mathbb{E}[\mathcal{D}_{\text{curv}}] \geq \frac{3}{8} \cdot q_h M_h/(8C_G) = \frac{3}{64}(q_h/C_G)M_h$. This proves (iii). □

Remark 39. Proposition 38 is deliberately minimal: one hidden layer, ReLU, bias variance $\sigma_b^2 > 0$ (which guarantees activation correlation across inputs; cf. the κ_2 discussion above). Its role is to certify that Theorem 14's two structural hypotheses are *theorems*, not merely assumptions, for at least one concrete member of each side of the pipeline—so the disparity law has a fully verified instance. For deeper heads and attention-based modules the hypotheses revert to assumption status, argued by signal propagation and checked empirically in Section 7.

Remark 40 (What the instance shows—and what it does not). The mechanism of Proposition 38 deserves a candid reading. The width-linear spike is generated by the *mean feature direction*: with $\sigma_b > 0$, every hidden unit has positive mean activation, so the features of all inputs share a common direction of $\Theta(M_h)$ energy. This requires no information in the features at all. Take every bottleneck feature $z_n = 0$, so that all inputs are indistinguishable: the features $h_n = \mathrm{ReLU}(b_1)$ are identical across pixels, carry no sample-specific structure whatsoever, and the proposition applies verbatim. Width-linear downstream curvature is therefore *not* evidence that the spatial module has found, or will find, useful structure: $\mathcal{D}_{\text{curv}}$ measures the downstream module's readiness to move—the loss curvature its parameters command at initialization—not the value of what it computes. This is the spatial-side counterpart of Remark 21: there, small spectral curvature cannot distinguish irrelevance from starvation; here, large spatial curvature cannot distinguish signal from capacity. It is also exactly why the geometry is a shortcut *hazard*: the capacity advantage is in place before the data has spoken, so whatever the spatial module can most easily fit—signal or shortcut—it is poised to fit first.

Remark 41 (Smoothness scope). ReLU is not twice differentiable, but no smoothness enters Proposition 38: the Gauss–Newton blocks involve only first-order Jacobians, which at Gaussian initialization exist almost surely (every preactivation is a.s. nonzero). Assumption 3 plays no role here—its function is the well-posedness of the gradient flow in Section 5, whose statements are formulated for absolutely continuous trajectories satisfying the flow almost everywhere and therefore accommodate piecewise-smooth models.

The other two conventions. The bound just proved is for the Gauss-Newton block $G_{\boldsymbol{\phi\phi}} = J_{\boldsymbol{\phi}}^\top H_\tau J_{\boldsymbol{\phi}}$ of (6), which is the object appearing in Definition 12. Two neighbouring matrices inherit it. For the *logit* Gram, $H_\tau \preceq I$ gives $J_{\boldsymbol{\phi}}^\top J_{\boldsymbol{\phi}} \succeq G_{\boldsymbol{\phi\phi}}$, so $\lambda_{\max}(J_{\boldsymbol{\phi}}^\top J_{\boldsymbol{\phi}}) \geq \lambda_{\max}(G_{\boldsymbol{\phi\phi}}) = \Omega(M)$ immediately. For the *residual* Gram $J_{\boldsymbol{\phi}}^{\mathrm{res}\top} J_{\boldsymbol{\phi}}^{\mathrm{res}} = J_{\boldsymbol{\phi}}^\top H_\tau^2 J_{\boldsymbol{\phi}}$, note that H_τ is symmetric with $H_\tau \mathbf{1} = 0$, so $\mathbf{1}^\perp$ is invariant and, by Step 2, $H_\tau^{(n)} \succeq p_{\min}^{(n)} I$ there; squaring the eigenvalues on that subspace gives $(H_\tau^{(n)})^2 \succeq (p_{\min}^{(n)})^2 P_\perp$, so the identical witness argument runs with the weights $(p_{\min}^{(n)})^2$ in place of $p_{\min}^{(n)}$ and yields $\lambda_{\max}(J_{\boldsymbol{\phi}}^{\mathrm{res}\top} J_{\boldsymbol{\phi}}^{\mathrm{res}}) \geq \lambda_{\max}(S_h^{p^2}) = \Omega(M)$ under the corresponding head hypothesis. All three conventions therefore satisfy Lemma 16, differing only in the softmax weighting carried by the feature Gram. This completes the proof. □

Remark 42 (Scope, and the role of the mean-field law). The witness argument uses only: (a) a dense classifier head over penultimate features of dimension $\Theta(M)$; (b) the head hypothesis $\lambda_{\max}(S_h^p) \geq q_h M_h$ at initialization—a *cross-input correlation* condition, which variance-scaled initialization does not by itself deliver (see the counterexample of per-vector norms above) but which holds for the standard activation choices discussed there; and (c) softmax cross-entropy with $N_{\text{cls}} \geq 2$. Given (b), the inequality $\lambda_{\max}(G_{\boldsymbol{\phi\phi}}) \geq \lambda_{\max}(S_h^p)$ is deterministic and architecture-agnostic, so the argument applies to per-pixel dense heads on MLP, CNN (channel count M), and ViT (embedding dimension M) spatial modules alike. For fully connected modules (b) follows from standard signal propagation at Kaiming initialization; for CNN and ViT heads we adopt it as an explicit assumption checked empirically. No i.i.d. Gaussian input assumption is required. Karakida et al.'s Theorem 4 plays the role of the sharp asymptotic benchmark: within its model class (fully connected, input and hidden widths proportional to M, output fixed at $C = \mathcal{O}(1)$, squared loss), it upgrades the witness lower bound to an exact width-linear law with explicit constant, and their Theorem 1 supplies the complementary fact that the *mean* eigenvalue shrinks as $\mathcal{O}(1/M)$, so curvature concentrates in a vanishing fraction of directions as the module widens. The loss transfer from squared loss to cross-entropy is handled exactly at initialization by Step 2, at the cost of the constant $p_{\min}$.

Remark 43 (Empirical status). Karakida et al.'s own numerical validation (their Figure 1, right column) confirms $\lambda_{\max}$ growing linearly in M, matching the exact constant of their Theorem 4, for tanh, ReLU, and linear fully connected networks. Our Experiment 1.1 tests the law on the SpatialMLP family, whose parameter count is *linear* in width ($C_g = \Theta(M)$, since input and output dimensions are fixed), so the width-linear law predicts a log–log slope of 1.0 against C_g as well as against M. The measured slope is 0.70 ± 0.05: the qualitative claim of Lemma 16—λ_ϕ grows without bound while $\lambda_{\boldsymbol{\theta}}$ stays capped—is confirmed unambiguously, but the measured exponent is sublinear, short of the width-linear asymptote. Candidate causes (finite width, the loss-Hessian versus Gauss–Newton measurement gap, correlated non-Gaussian features) are discussed in Section 7. We report the discrepancy rather than explain it away.

A.1.2 Proof of Lemma 18

The proof factors the spectral Jacobian via the chain rule and bounds each factor's operator norm using Assumptions 1, 4, and 7. We bound the *logit* Jacobian $J_{\boldsymbol{\theta}} = \partial \hat{y} / \partial \boldsymbol{\theta}$ of (3); the Gauss-Newton block and the residual Jacobian then inherit the bound, since both insert factors of H_τ with $0 \preceq H_\tau \preceq I$ (Step 4).

Step 1: Chain-rule factorisation. With $Z = f_{\boldsymbol{\theta}}(X)$ and $\hat{y} = g_{\boldsymbol{\phi}}(Z)$, the chain rule gives, in the normalization of (3) (the $N^{-1/2}$ riding on the right-hand factor, as fixed in Section 3),

$$J_{\boldsymbol{\theta}} \;=\; J_Z \cdot \frac{\partial Z}{\partial \boldsymbol{\theta}}, \qquad J_Z := \partial g_{\boldsymbol{\phi}}/\partial Z,$$

where J_Z is unnormalized—the object Assumption 7 bounds. Since $g_{\boldsymbol{\phi}}$ acts on one image at a time, the dataset-stacked J_Z is block diagonal across images, so its operator norm is the maximum over per-image blocks and Assumption 7 applies to the stack with the same constant $\widetilde{L}$. (For the residual map $r(\boldsymbol{\theta}, \boldsymbol{\phi}) = \tau(g_{\boldsymbol{\phi}}(Z))$ with $\tau(\hat{y}) = \mathrm{softmax}(\hat{y}) - y_{\mathrm{onehot}}$, the same computation carries the extra left factor $J_\tau := \partial\tau/\partial\hat{y} = \mathrm{diag}(p) - pp^\top = H_\tau$, the softmax Jacobian, which satisfies $\|H_\tau\|_{\mathrm{op}} \leq 1$; see Botev, Ritter, and Barber Botev et al. [2017].)

Step 2: Operator-norm inequality. Sub-multiplicativity of the operator norm gives

$$\|J_{\boldsymbol{\theta}}\|_{\mathrm{op}} \;\leq\; \|J_Z\|_{\mathrm{op}} \cdot \left\|\frac{\partial Z}{\partial \boldsymbol{\theta}}\right\|_{\mathrm{op}} \;\leq\; \widetilde{L} \cdot \left\|\frac{\partial Z}{\partial \boldsymbol{\theta}}\right\|_{\mathrm{op}},$$

using Assumption 7 on the first factor.

Step 3: The input-Jacobian factor, exactly. Under Assumption 1, $Z_n = W^\top x_n$ with $\boldsymbol{\theta} = \mathrm{vec}(W)$, $W \in \mathbb{R}^{S \times K}$. The per-pixel Jacobian is $\partial Z_n/\partial \mathrm{vec}(W) = I_K \otimes x_n^\top$ (Magnus and Neudecker Magnus and Neudecker [1988], Ch. 9). Stacking over the N pixel positions with the $N^{-1/2}$ normalization of (3) gives $\partial Z/\partial \boldsymbol{\theta} \in \mathbb{R}^{NK \times SK}$ with

$$\left(\frac{\partial Z}{\partial \boldsymbol{\theta}}\right)^{\top} \frac{\partial Z}{\partial \boldsymbol{\theta}} \;=\; \frac{1}{N}\sum_{n=1}^{N} (I_K \otimes x_n)\left(I_K \otimes x_n^\top\right) \;=\; I_K \otimes \Sigma_X,$$

hence, *exactly*,

$$\left\|\frac{\partial Z}{\partial \boldsymbol{\theta}}\right\|_{\mathrm{op}}^2 \;=\; \lambda_{\max}(I_K \otimes \Sigma_X) \;=\; \lambda_{\max}(\Sigma_X),$$

with Σ_X the mean per-pixel input Gram of (1); Assumption 4 bounds $\lambda_{\max}(\Sigma_X) \leq \Lambda_X$ uniformly in the dataset size N. Note that no factor of K appears: the Kronecker structure replicates the same spectrum K times rather than accumulating it.

Step 4: Assembling. Combining Steps 2 and 3,

$$\|J_{\boldsymbol{\theta}}\|_{\mathrm{op}}^2 \;\leq\; \widetilde{L}^2 \cdot \lambda_{\max}(\Sigma_X) \;=:\; C_{\boldsymbol{\theta}}.$$

For the Gauss-Newton block, $G_{\boldsymbol{\theta\theta}} = J_{\boldsymbol{\theta}}^\top H_\tau J_{\boldsymbol{\theta}} \preceq J_{\boldsymbol{\theta}}^\top J_{\boldsymbol{\theta}}$ since $H_\tau \preceq I$, so $\lambda_{\max}(G_{\boldsymbol{\theta\theta}}) \leq C_{\boldsymbol{\theta}}$; for the residual Jacobian, $\|H_\tau J_{\boldsymbol{\theta}}\|_{\mathrm{op}} \leq \|J_{\boldsymbol{\theta}}\|_{\mathrm{op}}$ gives the same cap. The constant $C_{\boldsymbol{\theta}}$ depends on the classifier's input Lipschitz constant $\widetilde{L}$ (which may grow with the depth of $g_{\boldsymbol{\phi}}$ but not with its width M; see the discussion of Assumption 7) and the top eigenvalue of the mean input Gram $\lambda_{\max}(\Sigma_X)$. It does not depend on M, on the spatial parameter count C_g, or on the dataset size N. This completes the proof. □

Remark 44 (Why an operator-norm cap suffices). An alternative route via the smallest positive eigenvalue $\lambda_{\min}^{+}(J_{\boldsymbol{\theta}}^\top J_{\boldsymbol{\theta}})$ requires either an exact Kronecker factorisation (valid only for per-pixel classifiers) or a trace-over-rank argument via the softmax residual dimension—both of which need additional structural assumptions on $g_{\boldsymbol{\phi}}$. The operator-norm bound above is significantly simpler, applies to general nonlinear spatial classifiers (CNN, ViT), and is exactly what Theorem 14 needs: a C_g-independent cap on the spectral block's *top* eigenvalue, so that $\mathcal{D}_{\mathrm{curv}} = \lambda_{\max}(G_{\boldsymbol{\phi\phi}})/\lambda_{\max}(G_{\boldsymbol{\theta\theta}}) \geq \Omega(M)/C_{\boldsymbol{\theta}}$.

Remark 45 (Softmax Jacobian bound). The bound $\|\mathrm{diag}(p) - pp^\top\|_{\mathrm{op}} \le 1$ is standard: $\mathrm{diag}(p) - pp^\top$ is the covariance matrix of the categorical distribution with parameter p (equivalently, the Jacobian of the softmax map), hence positive semi-definite with $\lambda_{\max} \le \max_i p_i \le 1$. See Botev, Ritter, and Barber Botev et al. [2017] for its role in Gauss–Newton computations.

A.1.3 Combining the lemmas: full proof of Theorem 14

Assemble Lemmas 16 and 18 to complete the proof. By Lemma 16, in expectation over random Kaiming (variance-scaled Gaussian) initialization,

$$\lambda_{\max}(G^{(M)}_{\boldsymbol{\phi\phi}}) \;\ge\; c_\phi \cdot M$$

for a constant $c_\phi > 0$ bounded below independently of M: the witness bound (12) gives $\mathbb{E}[\lambda_{\max}] \ge (1-\delta_h)\, q_h M_h \ge (1-\delta_h)\, q_h \beta M$, where $\beta > 0$ is the width-ratio floor $M_h \ge \beta M$ supplied by the head hypothesis, so one may take $c_\phi := (1-\delta_h) q_h \beta$. The witness route delivers this with no additive correction; within Karakida et al.'s Karakida et al. [2019] proportional-width model class, their Theorem 4 sharpens the width-linear rate to an exact asymptotic law. By Lemma 18,

$$\lambda_{\max}(G_{\boldsymbol{\theta\theta}}) \;\le\; \|J_{\boldsymbol{\theta}}\|^2_{\mathrm{op}} \;\le\; C_{\boldsymbol{\theta}}$$

with $C_{\boldsymbol{\theta}}$ independent of M and C_g. (If $C_{\boldsymbol{\theta}} = 0$ the spectral block vanishes identically and the bound holds trivially under the $+\infty$ convention of Definition 12; assume $C_{\boldsymbol{\theta}} > 0$ below.) The cap is deterministic—valid for every realization of the initialization—so it divides through inside the expectation:

$$\mathbb{E}\left[\mathcal{D}^{(M)}_{\mathrm{curv}}\right] \;=\; \mathbb{E}\left[\frac{\lambda_{\max}(G^{(M)}_{\boldsymbol{\phi\phi}})}{\lambda_{\max}(G_{\boldsymbol{\theta\theta}})}\right] \;\ge\; \frac{\mathbb{E}\left[\lambda_{\max}(G^{(M)}_{\boldsymbol{\phi\phi}})\right]}{C_{\boldsymbol{\theta}}} \;\ge\; \frac{c_\phi \cdot M}{C_{\boldsymbol{\theta}}} \;=\; c_1 \cdot M, \tag{13}$$

where $c_1 := c_\phi / C_{\boldsymbol{\theta}}$ depends only on the initialization scheme, the architecture class of $g_{\boldsymbol{\phi}}$, the classifier's input Lipschitz constant $\widetilde{L}$, the bottleneck dimension K, the input spectral dimension S, and the data distribution—exactly the dependencies claimed in the theorem statement. The witness route needs no additive correction ($c_2 = 0$ suffices); the subtracted constant is retained in the statement so that sharper asymptotic constants, which hold only for M large, may also be substituted. Under $C_g = \Theta(M^2)$, $M = \Theta(\sqrt{C_g})$ and the ratio form $\mathcal{D}_{\mathrm{curv}} \ge c_1' \sqrt{C_g/C_f} - c_2$ follows on multiplying and dividing by $\sqrt{C_f}$. This completes the proof. $\square$

Remark 46 (What Theorem 14 does and does not claim). Theorem 14 is a scaling law for the curvature ratio between two Gauss-Newton blocks; it does not claim that the full joint Gauss-Newton matrix has a large condition number in the strict $\lambda_{\max}/\lambda_{\min}$ sense (which is trivially infinite under rank-deficiency), nor that joint gradient descent produces disparate effective learning rates (which is the separate content of Theorem 27, under its own hypotheses). The scaling $\mathcal{D}_{\mathrm{curv}} = \Omega(M)$ (equivalently $\Omega(\sqrt{C_g/C_f})$ for $\Theta(M^2)$-parameter families) is a purely geometric statement about how curvature concentrates in the downstream module as the spatial network widens.

Remark 47 (Width scaling and the empirical exponent). The $\sqrt{C_g}$ scaling reflects the fact that Karakida's underlying result Karakida et al. [2019] (Theorem 4) gives $\lambda_{\max}$ in terms of the network's characteristic *width* M, not its total parameter count. For a fixed-depth architecture with $C_g = \Theta(M^2)$, the equivalent scaling is $\lambda_{\max}(J_{\boldsymbol{\phi}}^\top J_{\boldsymbol{\phi}}) \sim \sqrt{C_g}$ (log–log slope 0.5 against C_g). Experiment 1.1's SpatialMLP is *not* in this class: its parameter count is linear in width ($C_g = \Theta(M)$), so the applicable prediction there is slope 1.0 against C_g, and the measured 0.70 ± 0.05 is sublinear (Remark 43). The two parameter-count forms must not be conflated; the theorem's native variable is M.

A.2 Supplementary material for Theorem 27

The main-text proof of Theorem 27 is complete for the gradient-flow model (7): the displacement bound follows by direct integration of the residual envelope, with the operator-norm cap supplied by Lemma 18. The supplementary work concerns the hypotheses and the discrete extension.

Remark 48 (Open extensions). Three extensions are deliberately left open rather than sketched. (i) *Discrete SGD*: a Robbins–Monro tracking argument (Borkar Borkar [2008], Ch. 2) should carry the displacement bound to constant- and decaying-step SGD, with gradient noise entering the constants; we have not carried this out and state Theorem 27 for (almost-everywhere) gradient flow only. (ii) *Deriving the envelope*: obtaining Assumption 24—in particular the growth of μ with C_g—from the architecture, e.g. by an NTK analysis of the spatial pathway at large width adapting Pezeshki et al. Pezeshki et al. [2021], is the missing link that would chain Theorems 14 and 27 into a single causal statement; we test its content empirically (Experiment 1.2) instead of claiming it. (iii) *Envelope verification*: fitting $C/(1 + \mu t)$ envelopes to measured residual trajectories, with the fitted $\hat{\mu}$ increasing in capacity, is part of the experimental programme (Section 7).

A.3 Empirical robustness

The robustness studies referenced in the main text (optimizer, loss-variant, and regularizer ablations; the capacity-ratio threshold sweep) belong to the experimental programme of Section 7 and are reported there as they are run; this supplement makes no claim about them.